Government Debt and the Transmission of Monetary Policy to Credit

Elliot Spears[†]

January 2026

Abstract

This paper shows that government debt weakens the transmission of monetary policy to bank credit. Using a narrative approach to identify monetary policy shocks and using aggregated bank-level data, I find that contractionary policy has significantly smaller effects on lending and capital accumulation when debt-to-GDP is high. I explain this using a model with financial intermediaries in a fiscal theory framework with active fiscal and passive monetary policy. Higher debt raises inflation, dampening the response of real interest rates to policy tightening and attenuating its effects on credit. The model matches the empirical evidence and implies that sufficiently high debt counteracts the effects of monetary policy on credit. The model also predicts that debt maturity structure is relevant to monetary policy's transmission; as maturity structure shortens, monetary policy becomes even less effective under high-debt states at contracting credit.

[†] University of Washington, Department of Economics. Email: espear1@uw.edu.

1 Introduction

In the first half of 2024, the value of the United States' federal interest costs eclipsed military spending, making it the second largest source of federal outlays behind social security. Federal debt as a percent of GDP surpassed 100 for the first time in the nation's history in Q4 of 2012. The latest estimates put the federal debt-to-GDP in excess of 120%. As a result, economists have become increasingly concerned about the consequences of the U.S. fiscal trajectory. Mankiw (2025) delineated five ways out of the mounting debt: rapid growth, cutting spending, raising taxes, printing money, and default. Although AI and other technological advancements might generate significant growth, it's unlikely to generate the amount of resources necessary to get the debt under control. Cutting spending is nearly politically impossible. He cites the politically radioactive effects of even mentioning cuts to Social Security and Medicare; this will disincentivize legislators from making the necessary changes. Raising taxes was cited as the most politically feasible and least costly option between printing money and default.

However, history shows that countries that face high debt burdens have a tendency to experience frequent bursts of inflation. Under a healthy arrangement, the status quo will be for a central bank to predetermine the growth of base money. The central government's spending is then constrained by the public's willingness to purchase its debt, predetermined seigniorage, and tax revenue. The government adjusts its spending to the surrounding environment. The state of affairs is very different when the government determines how much it is going to spend irrespective of constraints. In this scenario, the central bank must finance, via money printing, the shortfall in government revenues from bond sales and taxes needed to cover its spending. This is precisely the point made in Sargent and Wallace (1981) when they say that, under some circumstances, monetary policy might not be able to control inflation, even in the long run.

These two regimes correspond to what is commonly referred to as active-monetary/passive-fiscal policy, and active-fiscal/passive-monetary environments, respectively. This verbiage originally comes from Leeper (1991). Countries that have high debt and consistently high inflation fall under the latter designation. In the case of the U.S., the concern is that our economy is becoming increasingly characterized by fiscal dominance [1], which will continue to exert inflationary pressure, even over the long run, unless something is done to reverse course.

Consequently, the fiscal theory of the price level (FTPL) has become an increasingly relevant framework for understanding the dynamics of the macroeconomy under mounting public debt. The purpose of this paper is to use FTPL to analyze the joint effects of monetary policy and fiscal policy, in an environment of fiscal dominance, upon credit issuance and capital accumulation. Much of the focus in FTPL modeling has revolved around macroaggregates, such as output, inflation, policy rates, discount rates, and government surpluses. However, very few papers have incorporated financial frictions into this model, and none, so far, have analyzed how monetary policy interacts with debt-to-GDP to shape the dynamics of credit.

In terms of macro aggregates, my results mirror those in Caramp and Feilich (2024) and Brandao-Marques et al. (2024) who find a negative joint effect between monetary policy and government debt, meaning that as the debt burden rises, the macro economy becomes less responsive to a given monetary policy shock. In my model, as the economy's indebtedness rises, the sensitivity of lending to a given monetary policy shock is dampened. The primary mechanism producing this result is the maturity structure of the government's debt. As a greater proportion of government debt is characterized by short-term debt, raising interest rates generates wealth effects that dampen contractionary repricing effects of long-term debt as well as standard New-Keynesian substitution effects. Firms' capital accumulation, which depends upon access to credit, follows the same dynamics as lending – a muted response to any given monetary shock as debt grows.

This intuition is consistent with findings in Bonfim et al. (2025) that show how the Portuguese government's procurement cuts during the 2010-2011 European sovereign debt crisis adversely affected firms that previously held government contracts, the majority of which were in construction. These cuts created a proliferation of non-performing loans (NPLs) that forced credit contractions. The converse of this behavior, where budget deficits

[1] A term for the scenario when fiscal policy "dominates" monetary policy, or an active-fiscal/passive monetary regime.

persist, creating mounting debt, can sustain lending even in the face of large losses on bank balance sheets due to the re-pricing of its debt holdings. Bottero et al. (2020) show that as bank balance sheets become increasingly exposed to sovereign debt, a given "sovereign shock"[2] generates ever larger contractions in credit. Therefore, the muted response in credit to debt shocks that I observed in the data and reproduce in the model suggests that U.S. debt still carries a fairly low risk premium, which leads inflation expectations to dominate the dynamics of credit and macro aggregates.

The empirical design is as follows: I build on the identification strategy of Caramp and Feilich (2024), who document dampening effects of debt on macro aggregates. I extend their framework in two directions: (i) examining bank-level credit responses, and (ii) embedding the mechanism in a structural model with a banking sector. I use the Romer and Romer (2004) exogenous shock specification to identify monetary policy surprises. I'm particularly interested in the responses of variables to contractions, which will be the focus of this paper. After purging the effects of macroeconomic news releases from the identified shocks, I split the data for U.S. federal debt into two categories: high debt versus low debt, which corresponds to periods when debt is one standard deviation above its mean value in the sample and when it's at or below the mean sample, respectively, for all observations. I then split the observations into the pre-QE and post-QE[3] time frames.

Using the method from Jordà (2005) to estimate local projections (LPs), I estimate the response of lending, controlling for various balance sheet variables and macro aggregates, to the Romer and Romer (2004) monetary policy shock in a state dependent fashion, where the states are high versus low debt in the pre and post-QE samples. The IRFs show that lending's response to monetary policy shocks under high debt is consistently lower than their response under low debt at every horizon over which the LPs are estimated; this is true for both the pre and post-QE samples. The main focus is in the pre-QE sample. The difference in lending under high versus low debt is as much as 60 basis points within a year of the shock.

To explain my findings, I build a medium-scale model with financial frictions that has

[2] A perceived increase in the riskiness of holding the sovereign's debt.

[3] Pre-QE: September 2008; Post-QE: March 2009 onward, where QE stands for Quantitative Easing

active fiscal policy and passive monetary policy. The fiscal block and calibrations for the Taylor rule are borrowed from Cochrane (2022a) and Cochrane (2022b). The financial block ties firm capital accumulation to its ability to obtain credit; that is, firms borrow from banks to accumulate additional capital. Then, I follow a similar pattern in developing the financial side of the economy to Balloch and Koby (2023), where bank loans and deposits are differentiated products sold by banks. Banks also hold government debt on their balance sheets and are therefore exposed to duration risk.

The predictions of the model are consistent with the empirical findings: for every 1% monetary policy contraction, each 1% rise in debt-to-GDP ratio reduces the impact of monetary tightening on lending by 0.05%. Given that firms finance capital accumulation via loans, the fall in capital accumulation is also dampened by rising debt. Counterfactual exercises show that as debt-to-GDP grows, a threshold is eventually reached where, even in the face of the monetary policy shock, lending and output increase as real interest rates begin to fall. For a 1% monetary policy shock, this threshold corresponds to about a 30% rise in debt-to-GDP.

Related Literature

This paper is at the intersection of public and macro finance. A broad literature exists that explores the effects of sovereign debt on the banking system. The Eurozone sovereign debt crisis of the 2010s was quite enlightening on this head. Altavilla et al. (2017); Bottero et al. (2020); Bonfim et al. (2025) are examples of work that map out the impact of sovereign debt risk on bank credit creation. A common thread is the classic doom-loop where risk of sovereign default generated a risk premium on the debt that pushed down bond prices. Banks that were exposed to this incurred capital losses and contracted credit, which generated further sovereign default risk.

Arellano et al. (2026) show that, in emerging markets, inflation expectations are highly sensitive to default risk and nearly 50% of inflation volatility in these countries is explained by default risk. Nakamura and Steinsson (2014); Auerbach et al. (2020) both argue that expansionary fiscal policy can have quite large effects on the real economy. The Auerbach et al. paper argues that this is also true of credit and that, in fact, we see the exact opposite of the crowding out effect. The mechanism is that fiscal expansions are income for firms, which raises their equity. When firm equity rises, default risk falls. As a result, the banks that lend to these firms experience asset quality improvement and they become better capitalized. Thus, credit expands, which pushes down borrowing costs. However, it is important to note that this paper is limited to the U.S. experience. Kosekova et al. (2025) show that there are fundamental differences across countries in terms of firm-bank relationships and that these relationships alter the transmission of policy shocks. Therefore, heterogeneous firm-bank relations across countries will likely lead to heterogeneous effects of debt on credit.

Very much related to my work is a paper by Cantero-Saiz et al. (2014) that uses evidence from Europe to conclude that monetary tightening is much more contractionary in high-sovereign-risk environments because banks face tighter funding constraints and cut lending more. As we shall see, my findings for the U.S. are almost the opposite. My explanation for this difference will point to the differences in maturity structure between the U.S. and the European countries.

My paper is also related to a large literature on government policy, both fiscal and monetary, and its transmission to credit through the banking system Gennaioli et al. (2014); Krishnamurthy and Muir (2017); Gennaioli et al. (2018). This is relevant because credit is intimately linked to investment. Cingano et al. (2016) show that a 10pp credit crunch reduces investment by 8-14%.

Finally, my model uses the FTPL framework. Its roots can be traced back to Sargent and Wallace (1981) and has been steadily developed over time Leeper (1991); Sims (2011); Bianchi et al. (2023); Cochrane (2023). As the U.S. debt burden becomes more of a concern, this framework will become more relevant. My contribution to this literature is to incorporate financial frictions into the model and exploit it to analyze the negative consequences that a rising debt burden has for monetary policy.

The remainder of the paper is structured as follows. In the next section (2.1) I lay out the empirical design. Then, in section (2.2) I discuss the results from the empirical analysis. In section (3.1) I turn to the quantitative model, providing a detailed overview of its ingredients. Section (3.2) displays the results of the simulation that are consistent with the empirical findings. Section (3.3) performs counterfactual analyses. Section (4) concludes.

2 Data

2.1 Monetary Policy Shocks

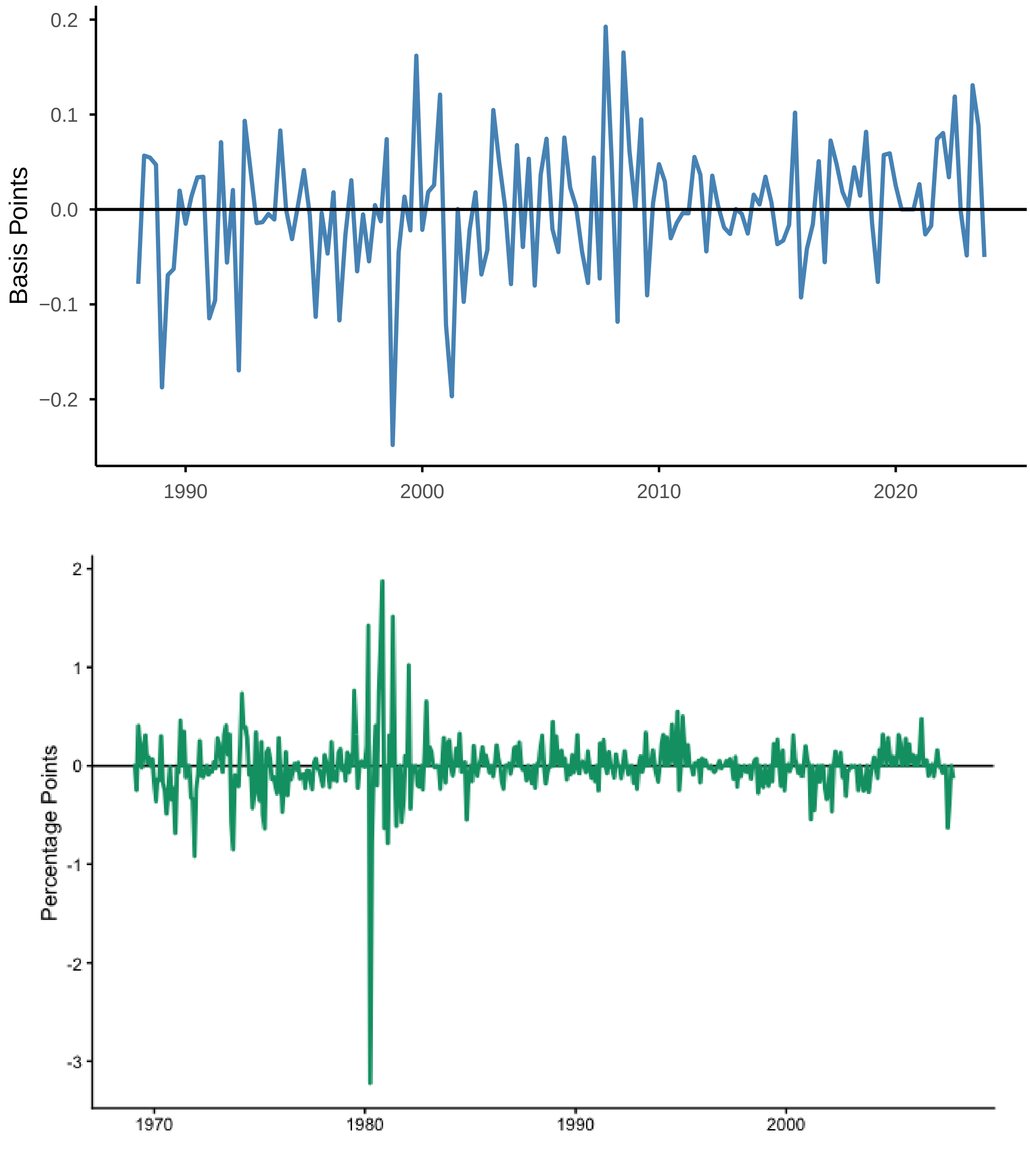


Figure 1: Quarterly Monetary Policy Shocks

Top: Bauer and Swanson (2023) orthogonalized monetary policy shock, aggregated to the quarterly level. Bottom: Romer and Romer (2004) monetary policy shocks, as updated by Wieland and Yang (2020). The orthogonalized series typically appears at a higher-than-quarterly frequency and is aggregated within quarter.

I begin with the identification of the monetary policy shock. There are two identification strategies that are employed widely in the literature to identify monetary policy surprises. The methodology that I will be relying on principally is the Romer and Romer (2004) identification. This shock series extends back to 1968 and, thanks to the work of Wieland and Yang (2020), has been updated continuously to the year 2008.

To pin down a monetary policy surprise, we need to eliminate any "anticipatory movements." The way Romer and Romer (2004) do this is by using internal memos and documents such as the Bluebook to pin down the change in the Fed's intended federal funds rate around each FOMC meeting. They then regress this measure on the Fed's Greenbook forecasts, which contain data on important macro aggregates (output, inflation, unemployment). The residual of this regression is the surprise to monetary policy that is not explained by forecasts. Therefore, it is taken to be our measure of a monetary policy shock.

For robustness tests as well as tests for the post-QE period, I use the monetary policy shocks from Bauer and Swanson (2023). The data for this shock is available through the San Francisco Federal Reserve Bank's Center for Monetary Research under the heading *Monetary Policy Surprises* (MPS). Figure 1 provides a visual of the series at quarterly intervals. Positive and negative values represent expansionary and contractionary policy shocks, respectively.

The methodology from Bauer and Swanson (2023) builds upon the work of Gurkaynak et al. (2005), where MPS is identified by following changes in the prices of futures contracts in a 30-minute window around FOMC meetings. Specifically, in the method employed here, the authors follow the changes in Eurodollar futures contracts 10 minutes prior to each FOMC meeting and 20 minutes after in order to measure the "surprise" behind FOMC announcements. Consequently, this data only extends to 2023 with the end of LIBOR. However, sources exist that use a variety of alternative securities to keep this series up to date.[4] The data begins in 1988Q1, and reaches its terminus at 2023Q4 with a total of 432 observations.

The final dataset that I use in the empirical analysis terminates in 2019Q4 due to the turbulence that followed in the wake of COVID-19. However, as a side note, the large

[4]SOFR futures have replaced Eurodollar futures for the purpose of updating this MPS series.

increase in government debt and the inflationary impact that ensued is what inspired this research. However, to be safe, I've chosen to omit this period.

The authors follow Nakamura and Steinsson (2018) in extracting a single principal component from this data that covers shocks to both the current federal funds rate target, as well as shocks to the future path of monetary policy (forward guidance). The shock is "orthogonalized" to various confounding factors, which means that the effects of macroeconomic news and other variables associated with changes in the futures contracts are regressed out of the shock measure, leaving us with a clean and well-identified monetary policy shock.

2.2 Bank Data

The main source of bank data comes from the H.8 commercial bank data release of the Board of Governors of the Federal Reserve. This source provides aggregate data on commercial bank lending, deposits, and other balance sheet variables of interest. The data spans from 1973Q1 - Present. However, due to difficulties in analyzing the effects of monetary policy surprises during the zero lower bound period, I will restrict our observations to the pre-QE period (2008). This means that our sample will run from 1973Q1 - 2007Q4. The frequency is monthly for all of the data, including the shocks, controls, and bank variables.

I perform some robustness checks in the appendix using more granular banking data that allows me to incorporate, among other things, bank-level fixed effects and controls for important bank balance sheet variables like capital ratios. The data on commercial banks was gathered using the entire universe of FFIEC Call Reports lending data available through Wharton Research Data Services (WRDS) spanning 1995Q3 - 2019Q4. The data occurs at quarterly frequency and contains about 643,000 observations on about 11,000 individual units over that interval. Some of the individual units do not span the entire time series, however, as some banks went out of or into business over that time. The same source allows me to also gather data on bank assets, tier 1 risk-based capital ratios, risk-weighted assets, deposits, and accumulated other comprehensive income (AOCI).

Given that the bank data is quarterly, while the MPS data is a higher frequency, I have to aggregate the shock to a quarterly measure. As other papers have noted Miranda-Agrippino (2016); Bauer and Swanson (2021), without controlling for the effects of macroeconomic news, this approach can lead to issues with endogeneity. I find that sticking with the controls[5] employed by Bauer and Swanson (2023) that orthogonalize the shock are sufficient for the purposes of quarterly aggregation of the shock. To verify that other sources of macro news don't contaminate the shock data, I regressed the quarterly-aggregated orthogonalized-MPS shock on a variety of macro news releases that precede each FOMC meeting, including, but not limited to: consumer sentiment, core inflation, retail sales (ex autos), and housing prices. This regression generated an $R^2 = 0.007$. Additionally, a joint F-test fails to reject that the macro news variables have no explanatory power for the already orthogonalized MPS shock ($F = 0.47,\ p \approx 0.80$), consistent with the interpretation of these shocks as orthogonal to standard macroeconomic news.

2.3 Public Debt Data

Figure 2 shows the evolution of U.S. treasury debt from the inception of the country to December 2025. This data is obtained from Hall and Sargent (2018) and will provide us with our measure of debt that we will use to construct our measure of debt-to-GDP. This measure is purged of its time trend before use in our analysis. The Hall and Sargent (2018) data occurs at a monthly frequency, which means we will have to gather monthly GDP data. I use the data from Stock and Watson (2010) to build the monthly debt-to-GDP measure.

The remaining data pertains to publicly available data obtained through the St. Louis Federal Reserve's Economic Data (FRED). This is comprised of quarterly data on Moody's Corporate Bond Yields (BAA10Y), GDP Deflator, and a measure of the unemployment gap constructed by subtracting the natural rate of unemployment from the actual unemployment rate over time.

3 Bank Lending, Debt, and Monetary Surprises

This section lays out the methodology used to generate the empirical results that shed light on the underlying question: how does credit respond to any given monetary policy

[5] There are six predictors of monetary surprises: nonfarm payrolls surprise, employment growth, S&P 500 index data, change in the yield curve's slope, Bloomberg commodity spot price index, and treasury skewness.

shock as debt-to-GDP grows?

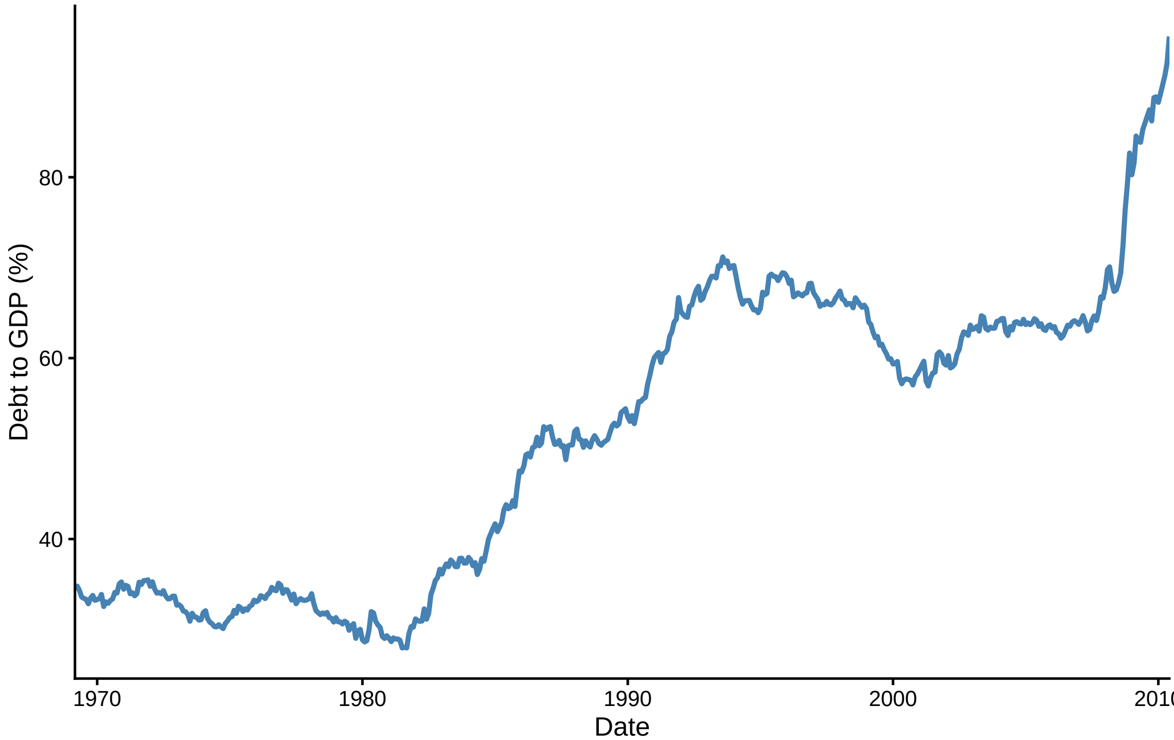


Figure 2: Gross Market Value of U.S. Treasury Debt (1968 - 2010)

Data from Hall and Sargent (2018). This plot displays the gross market value of U.S. treasury debt as a percentage of GDP at a monthly frequency. The monthly data for GDP is sourced from Stock and Watson (2010).

3.1 Methodology

When working with time-series data with the intention of analyzing how a macroeconomic variable of interest responds to shocks to other variables, one dominant approach follows the recommendations of Sims (1980) in using vector autoregression (VAR) models to generate impulse response functions (IRFs) of variables that are jointly determined. Although this is a standard approach in the macroeconomic literature, the appropriate way to tackle our question is by using Jordà's local projections (LPs) Jordà (2005). The reason is that we already have a well-identified exogenous shock pre-specified: our orthogonalized MPS shock. Given this external source of identification, the VAR framework is not required to recover structural shocks, and LPs provide a direct and flexible way to estimate impulse responses by projecting future outcomes on the shock of interest.

Under standard conditions, LPs and VAR-based impulse responses are asymptotically equivalent when the VAR is correctly specified. In practice, however, the two methodologies tend to differ in finite samples. VARs may be more statistically efficient when correctly specified (greater precision), while LPs are more robust to misspecification and more easily accommodate nonlinearities and state dependence. One tradeoff is that LP estimates tend to be less smooth and exhibit greater sampling variability at longer horizons, as in Hansen (2022).

I begin by showing how the method employed in this paper affects key macro aggregates. Figure 3 shows the interaction effect of debt with monetary policy shocks.

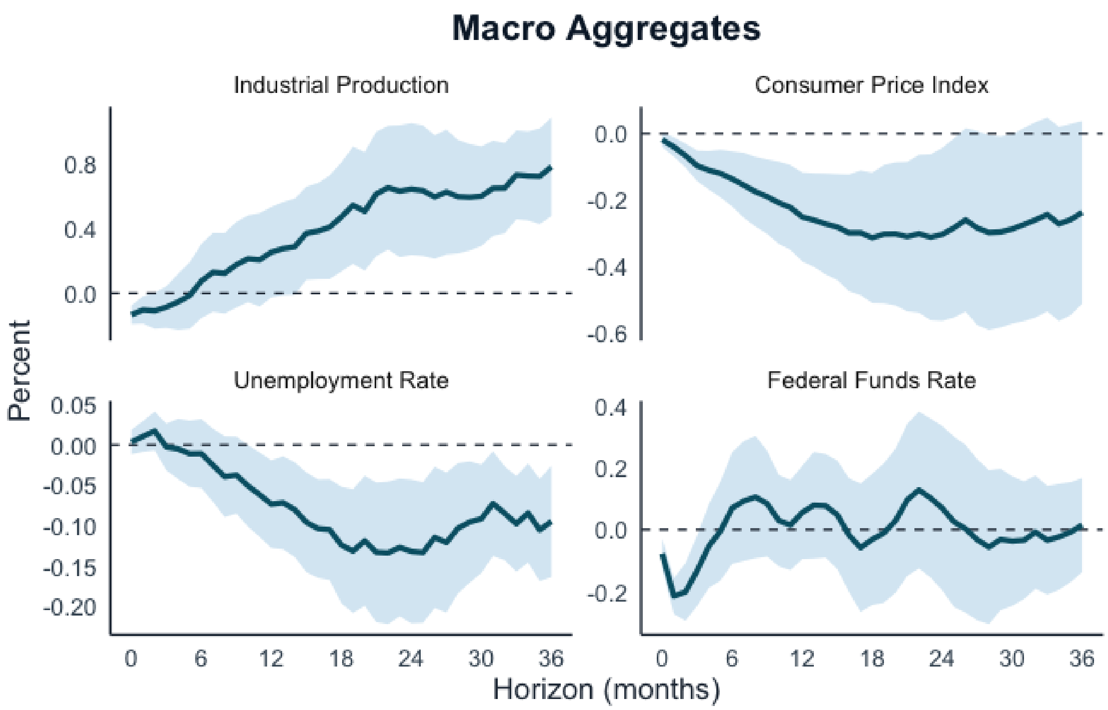


Figure 3: Baseline Test

Difference in the responses of macro aggregates to a 1 SD monetary policy shock under +1 SD debt minus mean debt.

3.2 Local Projections for Macro Aggregates

To estimate the dynamic effects of monetary policy shocks, I employ the local projection (LP) method of Jordà (2005). At each horizon $h = 0, 1, \ldots, H$, I estimate a separate regression of the form:

$$y_{t+h}^{(h)} = \beta_h \, \text{MPS}_t + \gamma_h \, D_{t-1} + \delta_h \, (\text{MPS}_t \times D_{t-1}) + \sum_{i=1}^{I} \phi_{h,i} \, \text{MPS}_{t-i} + \sum_{i=1}^{I} \boldsymbol{\theta}'_{h,i} \mathbf{X}_{t-i} + \varepsilon_{t+h} \quad (1)$$

where $y_{t+h}^{(h)}$ denotes the horizon-h outcome variable, MPS_t is the monetary policy shock, D_{t-1} is the lagged state variable capturing the level of government debt as a percentage of GDP (debt-to-GDP), and $\mathbf{X}_{t-i}$ is a vector of control variables. The government debt variable is lagged in order to allow the economy time to respond to changes in debt. This specification creates state dependence through the interaction of the monetary policy shock and debt-to-GDP $\text{MPS}_t \times D_{t-1}$.

The dependent variable is calculated in terms of its cumulative change between period $t-1$ and $t+h$. Specifically, for variables expressed in logs (e.g., industrial production or prices), we have:

$$y_{t+h}^{(h)} = \log Y_{t+h} - \log Y_{t-1}$$

while for variables in levels (e.g., unemployment or interest rates), I define:

$$y_{t+h}^{(h)} = Y_{t+h} - Y_{t-1}$$

The shock enters at time t, which means that this formulation implies that impulse responses should be interpreted as cumulative responses relative to the pre-shock period.

The variable MPS_t denotes the Romer and Romer (2004) monetary policy shock series in the main body of this paper. In the appendix, where I run robustness checks, the MPS_t term represents the Bauer and Swanson (2023) monetary policy shock.

The state variable D_{t-1} is the lagged debt-to-GDP ratio, standardized to have mean zero and unit variance. The vector $\mathbf{X}_t$ includes key macroeconomic controls: industrial production, the consumer price index, the producer price index, and the federal funds rate. The specification includes $I = 12$ lags of both the shock and all control variables.

Equation (1) is estimated separately for each horizon h using ordinary least squares. Standard errors are computed using Newey-West heteroskedasticity and autocorrelation consistent (HAC) estimators with a lag length equal to $h + 1$, allowing for increasing serial correlation at longer horizons.

The response at the mean level of debt ($D = 0$) is given by:

$$IRF_h^{\text{mean}} = \beta_h$$

At a higher level of debt (one standard deviation above the mean, $D = 1$), the response is:

$$IRF_h^{\text{high}} = \beta_h + \delta_h$$

The difference between high and average debt states is therefore:

$$IRF_h^{\text{diff}} = \delta_h$$

which captures the degree of state dependence in the transmission of monetary policy. As one can see in Figure 3, as we switch from an average-debt to high-debt state, the efficacy of monetary policy deteriorates. This is a contractionary monetary surprise, with unemployment rising by less and industrial production falling by less under the high-debt regime than under the low-debt regime.

3.3 Monetary Policy, Debt, and Lending

I estimate (1) once more, with a few changes. First the dependent variable will now be $y_t = 100 \cdot [\ln(\text{Loans}_t) - \ln(\text{CPI}_t)]$, which will be the real value of aggregate bank loans. The data for lending, as well as the rest of the data on commercial banks in this section, is obtained through the Board of Governor's release titled: H.8 Assets and Liabilities of Commercial Banks in the United States.

From the same commercial bank data, I add five additional variables to my set of controls in (1): securities (all), treasury securities, total assets, cash assets, and other assets. These controls are introduced in addition to the macro controls: industrial production, consumer price index, producer price index, federal funds rate, and the unemployment rate.

Instead of using $I = 12$ as in our baseline regression in the previous section, I use $I = 3$

for the control variables. I test for the optimal lag length using both the Akaike Information Criterion (AIC) and the Bayesian Information Criterion (BIC). AIC continues to decline as more lags are introduced all the way up to 12 lag lengths. Using a corrected AIC (AICc), it selects $H = 8$ when averaged across horizons. However, BIC penalizes additional lag lengths more heavily and ultimately chooses a more parsimonious model, selecting $H = 3$ when averaging across the first 12-month horizon and $H = 5$ over a 24-month horizon. I show the results of $H = 3$ in the appendix. Given the number of controls in the model, a concern about overfitting and a desire for parsimony lead me to stick with the $H = 5$ lag selection, given that we're stretching the IRFs over the longer horizon.

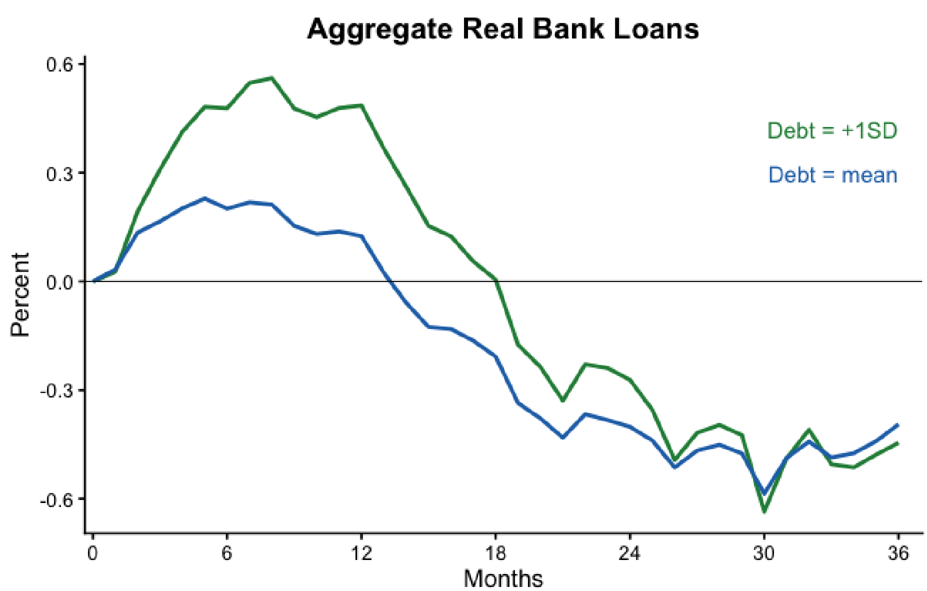


Figure 4: State-Dependent Loan Responses

Responses of aggregate commercial bank lending in the U.S. to a 1 SD monetary policy shock under +1 SD debt (green) versus mean debt (blue).

Figure 4 shows the resulting IRFs of lending in response to a contractionary monetary policy shock under the high-debt state and low-debt state. The green line corresponds to the response of commercial bank lending to a one-standard deviation monetary policy contraction under the high-debt state. The blue line does the same for the mean-debt state. The dynamics exhibit a classic hump shape in response to the shocks: at first we see a steady increase in lending in the wake of the contraction, before lending drops below trend after about a year and a half out.

The hump-shaped response of lending to a monetary contraction is a standard result seen elsewhere Bernanke and Blinder (1992); Nakamura (1988). This initial countercyclicality can be explained by a variety of factors, such as relationship lending Berger et al. (2024).

What is new is that debt appears to significantly dampen the contraction of credit. The peak of the initial positive hump more than doubles in magnitude when going from the mean to high-debt state. At longer horizons, after a year, the contraction is also less strong at every horizon. One possible explanation is what we observe in Figure 3, where aggregate activity contracts by less in the wake of monetary tightening under high debt.

As I mentioned above, we can also plot IRFs that represent the difference between the IRFs at each horizon. The IRF in Figure 5 shows the result of this exercise. This figure displays a statistically significant difference in lending behavior for over a year after the initial shock.

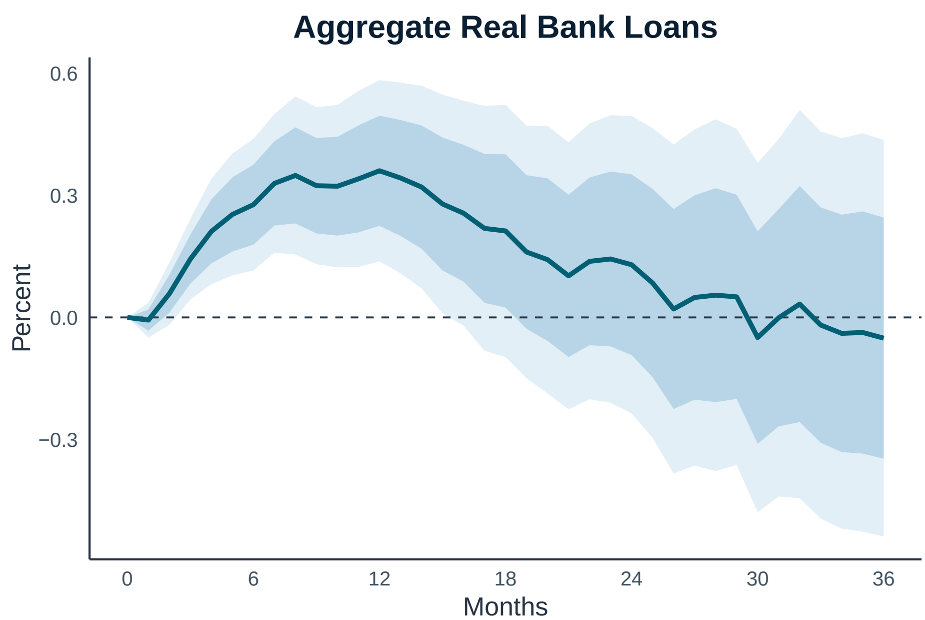


Figure 5: State-Dependent Loan Response Differential

State-dependent effects of a 1 SD monetary policy contraction of lending where the above IRF is generated by the difference between the high and low-debt states. Dark blue shades represent 68% confidence bands; light blue represents 90% confidence bands.

A clear pattern emerges from our plots: over the immediate near-term horizons, lending jumps up by more in the aftermath of the shock when the economy enters a high-debt state.

Although the initial increase in lending under mean-debt is more than corrected for after a year and a half by a persistent decline in lending thereafter, the initial jump in lending in the high debt state is not offset by a large and persistent fall over the medium to long term.

Therefore, my findings are consistent with Caramp and Feilich (2024) where I find a dampened effect on monetary policy as federal debt-to-GDP rises. Their results apply to macro aggregates, and my findings uncover a similar pattern with regards to credit. This contrasts with the findings of Cantero-Saiz et al. (2014) with respect to Europe. They find the opposite pattern: high debt entails greater contractionary effects on lending in the wake of a contractionary monetary policy shock.

One major difference that could serve as at least one explanation for this disparity is the maturity structure of the debt for various European countries and the sovereign risk tied to that debt. For instance, in Europe, around the time of the 2010 sovereign debt crisis, nearly 60% of debt had residual maturity under five years.[6]

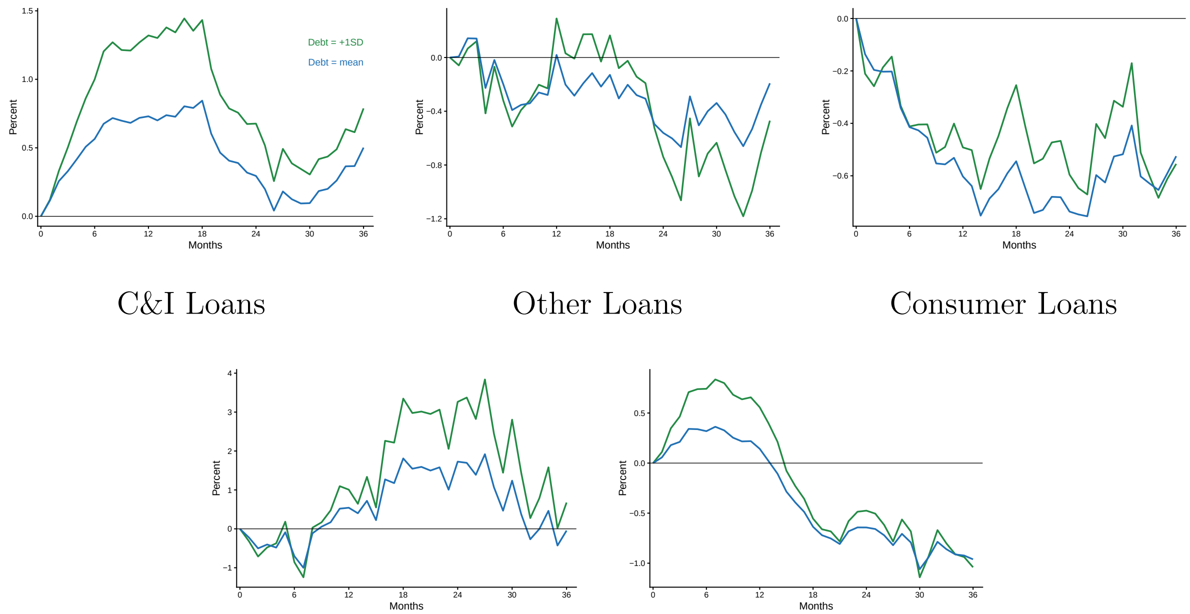
Another important fact is that, in the run up to the 2010 sovereign debt crisis in Europe, about 70% of Greece's foreign debt was owned by non-residents Lojsch et al. (2011) — especially German and French financial institutions. In the U.S. about two thirds of public debt are domestically held.[7] While sovereign risk became more of a concern for countries like Greece, who, unlike the U.S., cannot print off money to finance debt payments, the value of their debt instruments experienced larger oscillations than American debt. The disruptions this causes to holders of this kind of high-risk debt, which includes European creditors, may well explain the disparate findings between my results, those of Caramp and Feilich (2024) and the findings of Cantero-Saiz et al. (2014).

In Figure 6 I decompose the aggregate dynamics of bank lending in Figure 4. The resulting plot provides further confirmation that what is driving the initial increase in lending before eventually contracting is the commercial and industrial component. Consistent with Den Haan et al. (2007), consumer lending falls sharply and immediately, along with "other loans", which includes, for instance, loans to non-depository financial institutions; meanwhile

[6]The United States faces similar magnitudes of debt with residual maturity under five years Hall and Sargent (2018)

[7]U.S. Department of the Treasury, Bureau of the Fiscal Service. 2025. *Treasury Bulletin: Current Issue*. (September 2025).

C&I loans rise sharply. The explanation for this relates to the opposite behavior of "non-commitment loans" and "commitment loans" Morgan (1998), the latter of which expand after monetary tightenings as firms draw down on existing credit lines with their banks.



C&I Loans

Other Loans

Consumer Loans

Interbank Lending

Real Estate Loans (Commercial and Residential)

Figure 6: **Commercial Bank Lending by Category**

Notes: Plots of each lending category where green lines correspond to response to a monetary contraction at +1 SD debt and blue lines correspond to monetary contraction at mean debt.

Therefore, this evidence is consistent with the findings of Carpenter and Demiralp (2012) that lending is primarily demand-driven and that relationship lending ("commitment lending") can induce credit expansions in the wake of monetary contractions as firms increase their drawdowns.

Therefore, the initial *increase* in credit issuance that we observe in the wake of a monetary contraction in fig. 4 is not a bug of the modeling procedure — it is a real phenomenon; and as we see in that figure, as a country becomes increasingly indebted, that initial hump grows without a corresponding medium to long-run fall to correct for that expansion. This indicates that a nation's indebtedness, on the whole, dampens the response of credit to a given monetary policy shock.

3.4 Direct Federal Funds Rate Shock

A common thread in the literature is to assess the response of bank balance sheet variables to various shocks via a vector autoregression (VAR) model Bernanke and Blinder (1992); Bernanke and Gertler (1995). It has been of particular interest to see how these variables respond to a shock to the federal funds rate. So far, in this study, we've been identifying a monetary policy shock with a well-identified exogenous disturbance that plugs directly into the model. The most appropriate framework for analyzing the effects of this kind of shock, as previously discussed in section 3.1, is Local Projections.

Presumably, a positive (contractionary) monetary policy shock of the sort pinned down by Romer and Romer (2004) and Bauer and Swanson (2023) lead to an increase in the federal funds rate. [8] Therefore, it is of interest to test how lending responds to a direct increase in the federal funds rate and, for our purposes, how lending responds to this shock at varying levels of debt. The shock will represent a one standard deviation increase in the federal funds rate.

To this end, I specify the following VAR:

$$Y_t = \alpha + \Gamma D_{t-1} + \sum_{i=1}^{p} \beta_i Y_{t-i} + \sum_{i=1}^{p} \Phi_i (D_{t-1} \cdot Y_{t-i}) + u_t \tag{2}$$

Where Y_t is a vector of the endogenous variables: industrial production, CPI, unemployment rate, bank security holdings, bank treasury holdings, bank total assets, bank other assets, real loans, and the federal funds rate. The D_t term, once again, represents the standardized debt state variable, which equals 0 for debt at or below the mean level of debt in the sample, and equal to 1 for debt at or above one standard deviation above the sample mean. We have a mean-zero disturbance term u_t and intercept α.

The AIC and BIC are analyzed to select lag length p. AIC selects $p = 12$, while BIC selects $p = 1$. The resulting dynamics are not materially altered by one choice over the other. Therefore, I stick with the simpler choice dictated by BIC ($p = 1$). After estimating the VAR, the monetary policy shock, via the federal funds rate, is identified using a Cholesky

[8] Both of these papers confirm that their respective monetary policy shocks, in the event of a contraction, raise the federal funds rate.

decomposition. I then compute the regime-specific IRFs: the high and mean debt states.

It turns out that the VAR largely confirms the story that we get out of the local projections framework, but also provides additional nuance. The VAR's results overlap with the local projections' results in that, at early to medium horizons, which corresponds to a little over three years out, lending responds less to a given federal funds rate shock under the high debt regime than under the low debt regime. This is consistent with what we see in the local projections. This is shown in Figure 7. However, in the very long run, we see a reversal — lending falls more sharply after about three years under high debt than under low debt. This is the added nuance that the VAR provides.

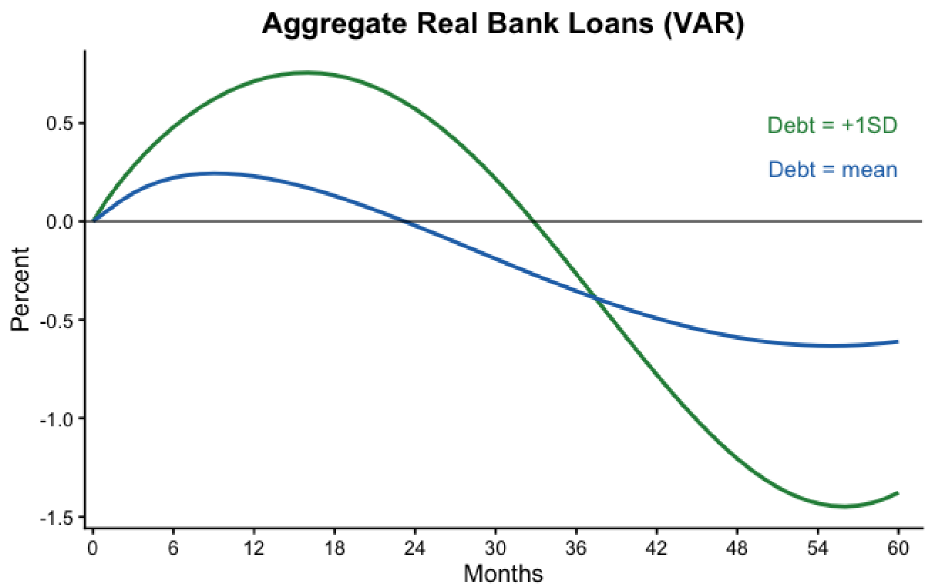


Figure 7: State-Dependent Loan Responses (VAR)

State-dependent effects upon commercial bank lending of a 1 SD shock directly to the federal funds rate.

Importantly, this says that, in the near to medium term, the effect of a contractionary shock to the federal funds rate on lending is muted by the presence of higher indebtedness. Only after several years does lending respond in the desired direction of a contraction, and at that point, it contracts more sharply than under average debt. The point here is that debt alters the transmission of monetary policy shocks vis-à-vis credit: lending dynamics, at every point, are amplified in one direction or another. At early stages, lending increases by more, and at later stages, it contracts more strongly. This trend is visualized by Figure 8 by looking at the differential of the lending dynamics at high versus low debt in the same way we did with the LPs exercise.

It is remarked elsewhere in the literature Cochrane (2022b) that, all else equal, it is more desirable to have smooth and drawn-out dynamics of economic variables in response to a policy shock than to have large oscillations. One of the virtues of long-term debt is that it can help buffer the effects of monetary and fiscal shocks and spread out the dynamics of inflation and output. However, when a country becomes more indebted and has to roll over greater quantities of its debt, we expect to see the precise dynamics given above: less control of monetary policy over the economy's dynamics, both in the short and long run.

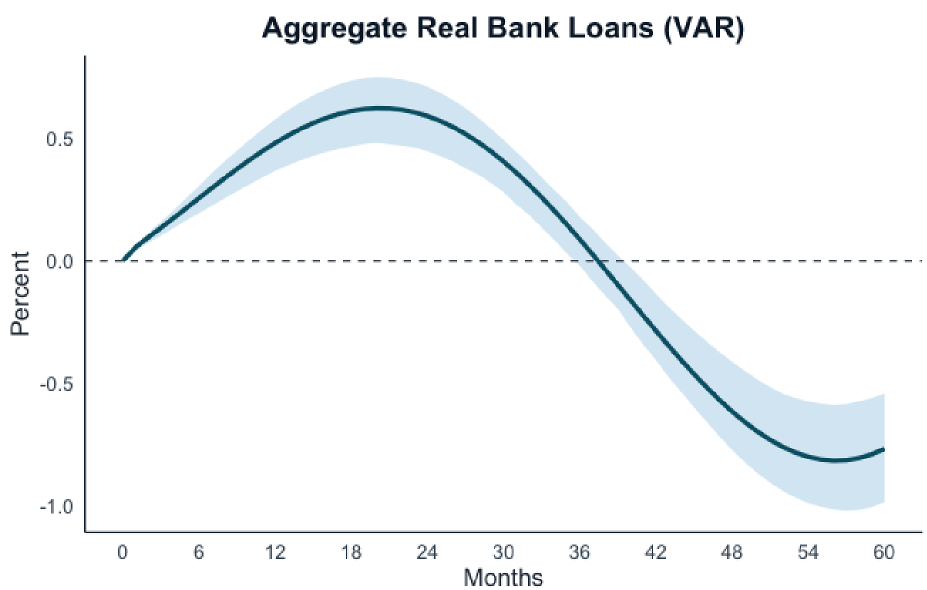


Figure 8: State-Dependent Loan Response Differential (VAR)

4 Fiscal Theory and Credit Model

The model operates in discrete time with an economy characterized by households with CRRA utility, final goods producers, and a bank that operates much like a final goods firm with CES aggregation of intermediate inputs. The firm side is standard, with one caveat: they use capital and labor to produce their good, while financing capital accumulation via loans from the bank. Banks operate in a fashion similar to the canonical intermediate-final goods pipeline in standard New Keynesian models.

The setup of the financial block largely follows Balloch and Koby (2023) in specifying loans and deposits as CES aggregates. There is a loan-services firm that aggregates intermediate bank loans according to a CES technology. Similarly, a savings product is sold by this firm in the form of a CES aggregate of bank deposits offered by the intermediate banks. Then, using Calvo frictions, I derive the rates at which loans and deposits are remunerated, which gives us spreads between the lending, deposit, and policy rates.

Finally, the model is characterized by active-fiscal/passive-monetary policy Leeper (1991). The skeleton of this block of the model is imported from Cochrane (2022a,b). The final model can ultimately be broken down into six equations and, in the appendix, I discuss the determinacy condition and how it contrasts with that of the log-linear three-equation NK model from Woodford (2003).

4.1 Households

Household preferences are given by:

$$\max_{C_t, N_t} \mathbb{E}_t \sum_{t=0}^{\infty} \beta^t \left[\frac{C_t^{1-\sigma}}{1-\sigma} - \frac{N_t^{1+\phi}}{1+\phi} + \psi \Phi \left(\frac{D_t}{P_t} \right) \right]$$

The household gets utility from consumption C_t and disutility from supplying labor N_t. They also receive some small utility from the liquidity services that deposits supply D_t with $\Phi'(\cdot) > 0$, $\Phi''(\cdot) < 0$. They maximize this utility, subject to the budget constraint:

$$P_t C_t + D_t + B_{s,t}^H + Q_t B_{\ell,t}^H = (1+i_t) B_{s,t}^H + (1+i_t^D) D_{t-1} + (1-\omega Q_t) B_{\ell,t}^H + W_t N_t + \Pi_{f,t} + \Pi_{b,t} + P_t T_t$$

Where $B_{s,t}^H$ is household holdings of short-term debt, which are remunerated at the policy rate i_t. The household also holds a portion of the government's issue of long-term debt $B_{\ell,t}^H$, priced at Q_t, with maturity structure governed by ω. They also receive government transfers T_t and, as owners of the bank and goods firm, they receive profits $\Pi_{f,t}$, $\Pi_{b,t}$.

4.2 Firms

The final good firm produces output that is a CES aggregate of intermediate goods:

$$Y_t = \left(\int_0^1 Y_t(j)^{\frac{\nu-1}{\nu}} dj \right)^{\frac{\nu}{\nu-1}}$$

Where $\nu > 1$ is the elasticity of substitution between intermediate goods. The final good firm's problem is therefore:

$$\max_{Y_t(j)} P_t \left(\int_0^1 Y_t(j)^{\frac{\nu-1}{\nu}} dj \right)^{\frac{\nu}{\nu-1}} - \int_0^1 P_t(j) Y_t(j) dj$$

Which, after taking the FOC and rearranging terms, eventually gives us:

$$Y_t(j) = \left(\frac{P_t(j)}{P_t} \right)^{-\nu} Y_t$$

We also have a continuum of intermediate goods firms $j \in [0, 1]$. Each firm produces a differentiated variety of output $Y_t(j)$ using capital K and labor N:

$$Y_t(j) = A_t K_t(j)^{\alpha} N_t(j)^{1-\alpha}, \qquad \alpha \in (0, 1)$$

where P_t is the output price, Y_t output, W_t the wage, N_t labor, K_t capital, L_t loans, i_t^L the gross nominal loan rate minus one, and δ the depreciation rate. Firms must finance their capital by borrowing from banks. Let $L_t(j)$ denote loans and $R_t^L \equiv 1 + i_t^L$ the loan rate. Firms are constrained in their ability to accumulate capital by how much they can borrow.

Assuming the borrowing constraint binds, profits become:

$$\Pi_t(j) = P_t(j) Y_t(j) - W_t N_t(j) - R_t^L K_t(j).$$

Given output $Y_t(j)$, the intermediate firms will seek to minimize total cost:

$$\min_{K_t(j),N_t(j)} W_t N_t(j) + R_t^L K_t(j) \quad \text{s.t.} \quad Y_t(j) = A_t K_t(j)^{\alpha} N_t(j)^{1-\alpha}.$$

Each period, a firm can reset its price with probability $1-\theta$. If not, its price remains fixed. A firm that can reset at time t chooses P_t^* to maximize expected discounted profits:

$$\max_{P_t^*} \sum_{k=0}^{\infty} (\beta\theta)^k \mathbb{E}_t \left[\Lambda_{t,t+k} \big(P_t^* Y_{t+k|t} - MC_{t+k} Y_{t+k|t} \big) \right], \tag{3}$$

where $\Lambda_{t,t+k}$ is the stochastic discount factor and

$$Y_{t+k|t} = \left(\frac{P_t^*}{P_{t+k}} \right)^{-\nu} Y_{t+k}.$$

This comes from the solution to the final goods producer's problem. $Y_{t+k|t}$ is the output of the firm k periods ahead of the last time they reset their price at time t. The first-order condition of (4) yields the optimal reset price:

$$P_t^* = \frac{\nu}{\nu-1} \frac{\sum_{k=0}^{\infty} (\beta\theta)^k \mathbb{E}_t \left[\Lambda_{t,t+k} P_{t+k}^{\nu} MC_{t+k} Y_{t+k} \right]}{\sum_{k=0}^{\infty} (\beta\theta)^k \mathbb{E}_t \left[\Lambda_{t,t+k} P_{t+k}^{\nu-1} Y_{t+k} \right]}.$$

Under Calvo pricing, log-linearization around a steady state yields the New Keynesian Phillips Curve (NKPC):

$$\pi_t = \beta \mathbb{E}_t[\pi_{t+1}] + \kappa \, mc_t, \tag{4}$$

4.3 Banks

Total lending is a CES aggregate of a continuum of banks:

$$L_t = \left(\int_0^1 L_t(j)^{\frac{\varepsilon-1}{\varepsilon}} dj \right)^{\frac{\varepsilon}{\varepsilon-1}}$$

That is, a competitive loan-services firm aggregates intermediate bank loans according to a CES technology, much like the Calvo setup for final goods firms. The idea is that loans are differentiated products sold by banks and ε is the elasticity of substitution between loans as in Balloch and Koby (2023). The profit maximization problem of the loan-services firm is:

$$\max_{L_t(j)} R_t^L \left(\int_0^1 L_t(j)^{\frac{\varepsilon-1}{\varepsilon}} dj \right)^{\frac{\varepsilon}{\varepsilon-1}} - \int_0^1 R_t^L(j) L_t(j) dj$$

Taking FOCs and simplifying, we get the demand for each intermediate (j), which is proportional to total loans, L_t:

$$L_t(j) = \left(\frac{R_t^L(j)}{R_t^L} \right)^{-\varepsilon} L_t$$

We also get an expression for the aggregate lending rate:

$$R_t^L = \left(\int_0^1 R_t^L(j)^{1-\epsilon} dj \right)^{\frac{1}{1-\varepsilon}}$$

Intermediate banks experience nominal profits:

$$\Pi_t(j) = R_t^L(j) L_t(j) + R_{t+1}^n Q_t B_t - R_t^D(j) D_t(j)$$

Where they are remunerated for their loans at rate $R_t^L(j)$. They also hold government debt B_t with a market price Q_t and expected period-by-period return R_{t+1}. Banks remunerate deposits at rate $R_t^D(j)$ and, along with equity $E_t(j)$, use these deposits to finance the accumulation of loans and government debt. Banks are also subject to the capital constraint:

$$L_t(j) \leq \eta E_t(j)$$

and they are also subject to the following budget constraint:

$$L_t(j) + Q_t B_t(j) = D_t(j) + E_t(j)$$

When banks set $R_t^L(j)$ so as to maximize profits, and after simplifying and rearranging conditions, we end up with the following condition for the loan rate:

$$R_t^L(j) = \underbrace{\frac{\varepsilon}{\varepsilon - 1}\left(1 - \frac{1}{\eta}\right)}_{\text{markup}} R_t^D(j).$$

Where, ε and $\eta > 1$. Therefore, we have a wedge between the loan and deposit rates that represents the markup banks charge for their loans over and above what they remunerate their depositors.

4.4 Government and Central Bank

Following the notation of Cochrane (2022a), the government, at each period t, faces nominal market value of debt V_t, which is defined as:

$$V_t = M_t + \sum_{j=0}^{\infty} Q_t^{t+1+j} B_t^{t+1+j}$$

Where M_t is nominal money balances in the economy at time t. The component of debt that drives the dynamics of the model is the term $Q_t^{t+1+j} B_t^{t+1+j}$, where B_t^{t+j} is the nominal amount of zero-coupon debt at period t that matures at period $t+j$. Q_t^{t+j} is the corresponding price of this debt.

Earlier, in the bank's problem, we saw that government bonds earn an expected nominal return R_{t+1}^n. This is entirely determined by the difference in the price of bonds Q_t overnight:

$$R_{t+1}^n = \frac{M_t + \sum_{j=1}^{\infty} Q_{t+1}^{t+j} B_t^{t+j}}{M_t + \sum_{j=1}^{\infty} Q_t^{t+j} B_t^{t+j}}$$

Bonds have a geometric maturity structure determined by $0 < \omega < 1$, which gives $B_t^{t+j} = \omega^{j-1} B_t$. As ω increases, the maturity structure is characterized by more long-term debt. As we shall see in our experimentation with the model, as we shorten the duration of debt, negative surplus shocks (or positive debt shocks), generate more inflation. The idea is that, with more short-term debt, the government is more exposed to rollover risk, which affects discount rates today and, furthermore, inflation.

The government's fiscal policy is set so as to raise surpluses to, at least partially, repay accumulated debts:

$$s_t = \theta_{s\pi}\pi_t + \theta_{sx}y_t + \alpha_v v_t + u_t^s \tag{5}$$

Where lowercase v_t is log market value of debt-to-GDP and u_t^s is an exogenous disturbance term. That surpluses don't simply respond one-to-one to unexpected changes in inflation will give us "active" fiscal policy. The central bank has a policy rule that simply follows the standard Taylor rule:

$$i_t = \theta_{i\pi}\,\pi_t + \theta_{iy}\,y_t + u_t^i \tag{6}$$

Where u_t^i is an exogenous disturbance. Unlike the standard New Keynesian setup, we will set $\theta_{i\pi} < 1$, so that interest rates respond less than one-to-one to inflation, hence the "passive" monetary policy. The remaining details of the model are laid out in the appendix, including the calibration of parameters, which can be found in Table 5. The calibration of the monetary and fiscal policy rules are taken from Cochrane (2022b), with banking parameters calibrated to match interest rate spreads.

The model simulations will involve negative shocks fed through u_t^s in equation (5), which are equivalent to an unbacked fiscal expansion. They will also, simultaneously involve a positive shock to equation (6) through u_t^i, which represents a monetary contraction. Figure 9 illustrates the competing effects these two shocks have on inflation. As governments fail to generate sufficient future surpluses to cover past deficits, this creates inflationary pressure through a variety of mechanisms (wealth effects, flight into goods, etc.), which counteract the deflationary effects of a monetary contraction. With long-term debt, these inflationary effects can be pushed forward into the future and smoothed out, which means the inflationary pressure of deficits can outlast the deflationary effects of monetary contraction.

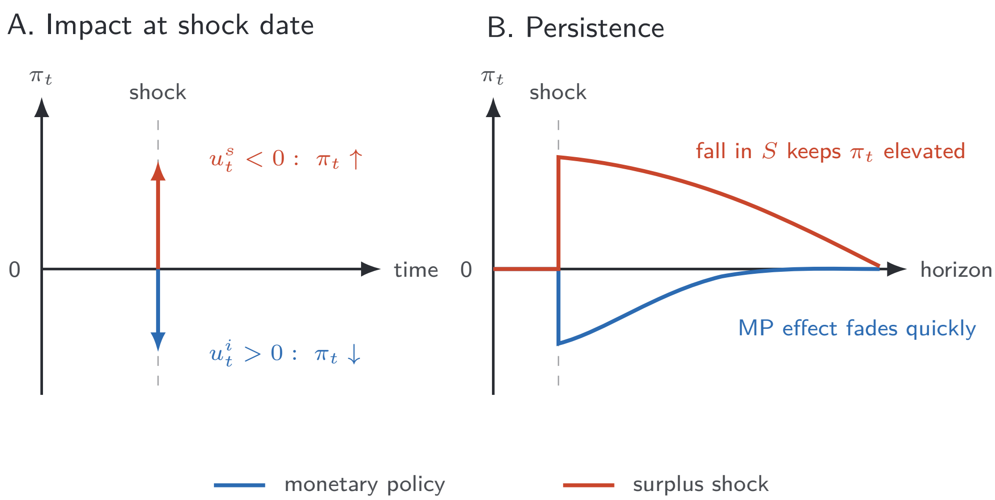


Figure 9: Joint Shocks and Their Competing Effects on Inflation

This figure shows the competing effects that a simultaneous negative surplus S shock $u_t^s < 0$ (equivalently, an unfunded fiscal expansion) and contractionary monetary policy MP shock $u_t^i > 0$ can have on inflation π_t.

4.5 Parameterization

In calibrating the model, I split the parameters into two sets: a set of pre-selected, fixed parameters; and a set of parameters chosen to match empirical moments in my previous findings.

Table 1: Baseline Calibration

Parameter	Value	Description	Source
$\theta_{i\pi}$	0.80	Taylor rule response to inflation	Cochrane (2022b)
θ_{iy}	0.50	Taylor rule response to output	Cochrane (2022b)
θ_{sy}	1.00	Fiscal surplus response to output	Cochrane (2022b)
η	10	Bank leverage constraint parameter	Balloch and Koby (2023)
ε	7	Loan demand elasticity	Balloch and Koby (2023)

Notes: This table shows values of the pre-selected parameters and their corresponding sources.

The key parameters of interest are the calibrations for the Taylor and surplus rules. One of the key choices is $\theta_{i\pi} = 0.8$, which in standard New-Keynesian models, is usually calibrated around 1.5. This latter choice would be consistent with an "active money" framework; our choice gives us passive money. The interest rate response to output $\theta_{iy} = 0.5$ is standard. Cochrane (2022b) calibrates the surplus rule to respond less than one-to-one to debt $\alpha_v = 0.20$ and inflation $\theta_{s\pi} = 0.25$, giving us an "active fiscal" environment; however, I estimate these using data in the procedure outlined below. The parameter η determines the bank's capital requirement and is taken from Balloch and Koby (2023) along with the loan demand elasticity ϵ.

Table 4 in the appendix provides an overview of moments in the data used to determine the magnitudes of the deviations from trend that each shock represents. I use these magnitudes to set the size of each shock that will be used in our moment-matching procedure. Here, I match two key moments in the data: the state-dependent effects of a monetary contraction on lending, and the state-dependent effects on the real interest rate. In both cases I estimate the state-dependent effect of a monetary policy shock in the model ($state^{mod}$):

$$state^{mod} = \frac{1}{36}\sum_{h=1}^{36} IRF_h(X_{high}) - IRF_h(X_{mean}) \tag{7}$$

Where $IRF_h(X_{high})$ is the impulse response of variable X (loans, or real rate) in the quantitative model. The IRF for the real interest rate can be found in Figure 17 in the appendix.

Recall from section 3.2 the procedure used to generate the differential IRFs between the high debt state and the low debt state. The object of interest is $IRF_h^{\text{diff}} = \delta_h$. Therefore, for both lending and real rates I'm interested in computing the local-projections equivalent of equation (7), which is the state-dependent effect of a monetary policy shock on each variable in the data ($state^{data}$):

$$state^{data,i} = \frac{1}{36}\sum_{h=1}^{36} \delta_h^i \tag{8}$$

for variable $i \in$ (loans, real interest rate). I set a range of possible values that each parameter of interest could possibly take on based upon previous and adjacent literature. I then build an algorithm that performs a grid search to minimize the distance between the two moments of interest discussed above. In gathering data on the empirical moments, I include confidence intervals at the 68% level.

Table 2: Empirical Targets

Moment	Data (%)		Model (%)
	Target	CI	
$state^{data,i}$			$state^{mod}$
i = Loans	0.1529	$[-0.0302 \ \ 0.3359]$	0.1375
i = Real Interest Rate	-0.0131	$[-0.0438 \ \ 0.0175]$	-0.0437

Notes: This table shows the empirical moments and the corresponding model estimates. For empirical moments under $state^{data,i}$, we estimate equation (8) for each variable i under "Target" and the corresponding 68% confidence bands under "CI". The model counterpart under $state^{mod}$ estimates equation (7) for each variable in the model.

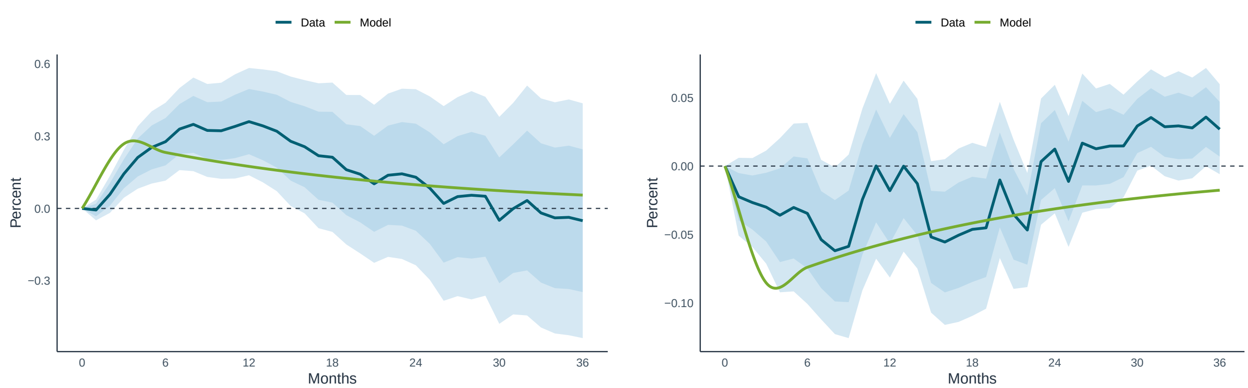
Table 2 shows the estimates of the average value of the IRF lines for Loans and the Real Interest Rate over the 36 period horizon, as estimated from the data and according to equation (8). This data appears under the "Target" column. The corresponding 68% confidence intervals are reported under "CI". The model fit, following the moment matching procedure, is reported in the column underneath $state^{mod}$. Table 3 reports the fitted values of the parameters that are generated by this procedure.

The fit of the model to the data is illustrated in Figure 10 below. The green lines represent the IRFs of the model after calibrating the shocks to match the magnitudes of the shocks in the data according to Table 4 and after fitting the model parameters in Table 3 to match the empirical moments, as displayed in Table 2.

Table 3: Fitted Parameters

Parameter	Range	Fitted Value	Description
σ	[1, 3]	2	Risk aversion
ρ_i	[0.1, 0.95]	0.2	MP shock persistence
ρ_s	[0.1, 0.95]	0.5	Fiscal shock persistence
ρ	[0.5, 0.95]	0.622	Debt valuation parameter
β	[0.96, 0.995]	0.97	Discount factor
κ	[0.05, 1]	0.25	NKPC slope
ρ_a	[0.7, 0.99]	0.85	TFP persistence
ω	[0.3, 0.99]	0.9	Maturity structure
α_v	[0.05, 0.50]	0.1	Surplus-debt rule
$\theta_{i\pi}$	[0.10, 0.50]	0.10	Inflation rule

At this point, I turn to an exercise that will quantify the effects of rising debt on monetary policy's efficacy in contracting credit. I will also examine the effects of changes to the maturity structure ω of government debt upon these same dynamics.



(a) Lending (b) Real Interest Rate

Figure 10: Model Fit

This figure illustrates the fit between the model and the data. The blue IRF represents data, while the green IRF represents the model's IRF counterpart. Dark and light bands represent 68% and 90% confidence bands, respectively.

4.6 Monetary Policy Shocks and Debt

The question of interest is how credit responds to a given monetary policy shock as a country becomes more indebted. To be clear, the kind of economy I am interested in simulating is one that issues nominal debt in its own currency and, therefore, has the ability to inflate away existing debt. Our model operates under that exact framework. Our model has a government that issues long-term debt that it partially repays with future surpluses, thereby generating transitory inflation when debt deviates from trend.

We can use our model to simulate the dynamics of credit in the wake of a monetary policy shock, given varying deviations of debt-to-GDP from its long-term trend. The monetary policy shock represents a one percentage point increase in i_t (MP contraction), which enters through the Taylor rule (7). I simultaneously force surpluses to deviate from trend by four different degrees, from smallest to largest: 1 pp, 5 pp, 10 pp, and 20 pp.

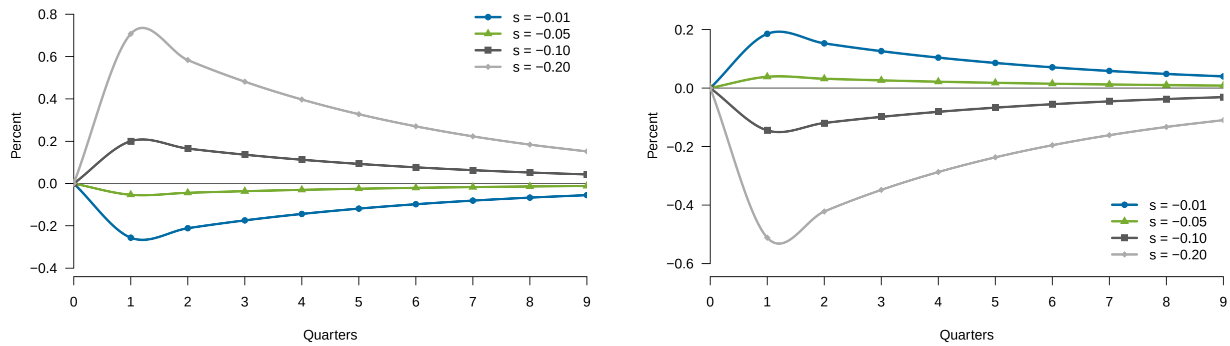
One might wonder why I'm introducing shocks to surpluses instead of debt directly. Recall that the valuation equation specifies the following relation:[9]

$$\frac{M_{t-1} + \sum_{j=0}^{\infty} Q_t^{t+j} B_{t-1}^{t+j}}{P_t} = \mathbb{E}_t \sum_{j=0}^{\infty} \beta^j s_{t+j} \tag{9}$$

Here, the price level adjusts so that the real value of nominal government debt (LHS) equals the present value of future surpluses (RHS). Consider an exogenous rise in the value of debt (LHS) through B_{t-1}. Barring a corresponding rise in surpluses, the price level must rise or bond prices Q_t must fall, or both, to reestablish equality. Likewise, if surpluses fall relative to the LHS, then prices must rise or bond prices fall or both. A negative surplus shock gives us large deficits, which raise the value of debt.

Figure 11 illustrates the responses of lending and the real interest rate to varying degrees of deviation of debt from trend. Both panels face a 1% monetary policy contraction. Each line represents a different magnitude of unbacked fiscal expansion, with the blue line representing the smallest unbacked expansion (1% fall in future expected surpluses), while the gray line represents the largest unbacked expansion (20% drop in future expected surpluses).

[9] See Cochrane (2023), page 53.



(a) Lending

(b) Real Interest Rate

Figure 11: Joint Shock Impulse Responses with Varying Debt Shock Magnitudes

Both panels face a 1% monetary policy contraction. (a) shows IRFs for lending at varying deviations of expected future surpluses from trend. (b) does the same for the real interest rate.

The results are consistent with the empirical predictions and the proposed mechanism — for a given monetary policy contraction, as a nation's indebtedness rises relative to the resources it has available to meet its obligations, both lending and real interest rates respond by less to a given monetary contraction. As indebtedness increases, the differential between real rates under the high-debt state and mean-debt state eventually slides below 0, leading to an increasingly large gap between lending responses to monetary contractions under high debt versus its response under mean debt.

Recall that our model has geometric maturity structure ω, which the moment matching procedure calibrated to be $\omega = 0.9$, which is consistent with the existing literature Cochrane (2022a,b). This means that as we decrease ω, we shorten the maturity structure. A shorter maturity structure increases the frequency with which debt must be rolled over, which compresses inflation dynamics.[10]

As we gradually shorten the maturity structure we see that the ability of monetary policy to contract credit when necessary becomes increasingly limited in the face of high debt. This tracks with our expectations since shortening maturity structure leads to larger inflationary spikes as debt deviates from trend, which generates larger effects on real interest rates, which alter the path of lending. Therefore, maturity structure of debt is also an important factor

[10] Having more long-term debt (higher ω) smooths out the inflation dynamics and pushes inflation forward in the wake of fiscal shocks.

that can impede the goals of monetary policy as a country becomes more indebted.

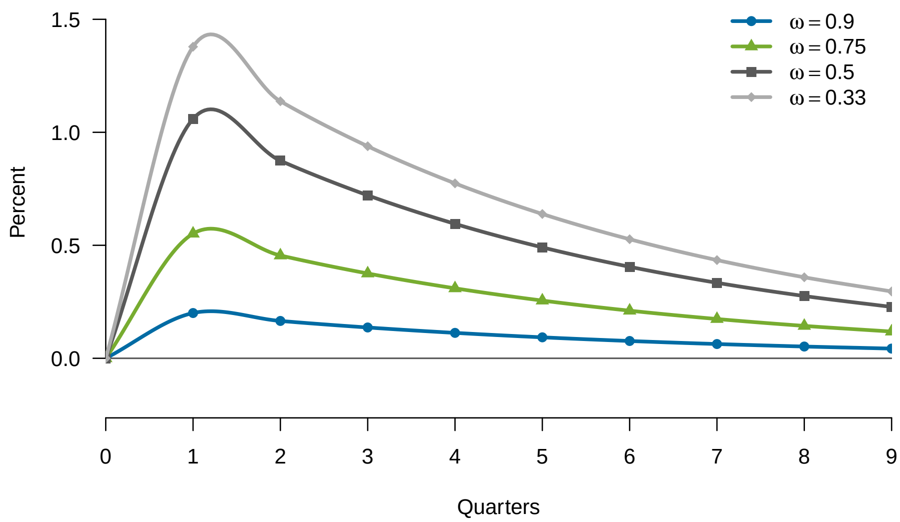


Figure 12: Lending IRFs by Maturity Structure

This figure shows the response of lending to the standard joint -28% surplus shock and 0.3% monetary contraction laid out in Table 4 under varying debt maturity structures ω

5 Conclusion

This paper set out to analyze the joint effects of monetary and fiscal policy in the United States. The data shows that as the country becomes more indebted relative to its available resources, a given monetary policy shock contracts credit growth by less. This operates through the path of inflation and, furthermore, real interest rates. As a country's fiscal outlook deteriorates, individuals expect the government to respond by inflating away at least a portion of the debt. As the debt grows, with no change in expected future fiscal backing, the contractionary effects of monetary policy on real interest rates are dampened, which, furthermore, causes credit to contract by less.

Using a model with active fiscal and passive monetary policy, I match empirical moments before showing that debt's maturity structure is also a relevant factor to the dynamics of credit in the wake of monetary tightening; as a country issues more short-term debt and shortens its maturity structure, in the face of unbacked fiscal expansions, debt-rollover frequency will increase, generating more inflation and further dampening credit dynamics in the wake of a monetary contraction.

The implications of this are clear: in an environment of fiscal dominance, fiscal policy has at least as much a role to play in determining the path of credit and inflation over the short and long term as monetary policy.

References

Altavilla, C., M. Pagano, and S. Simonelli (2017, October). Bank Exposures and Sovereign Stress Transmission*. *Review of Finance 21*(6), 2103–2139.

Arellano, C., Y. Bai, and G. Mihalache (2026). Monetary policy and sovereign risk in emerging economies (nk-default). *The Quarterly Journal of Economics*, qjag012.

Auerbach, A. J., Y. Gorodnichenko, and D. Murphy (2020, May). Effects of Fiscal Policy on Credit Markets. *AEA Papers and Proceedings 110*, 119–124.

Balloch, C. and Y. Koby (2023). *Low rates and bank loan supply: Theory and evidence from Japan*. LSE Financial Markets Group.

Barnichon, R. and C. Brownlees (2019). Impulse response estimation by smooth local projections. *Review of Economics and Statistics 101*(3), 522–530.

Bauer, M. D. and E. T. Swanson (2021). The fed's response to economic news explains the "fed information effect". Technical report, IMFS Working Paper Series.

Bauer, M. D. and E. T. Swanson (2023). A reassessment of monetary policy surprises and high-frequency identification. *NBER Macroeconomics Annual 37*(1), 87–155.

Ben Zeev, N., V. A. Ramey, and S. Zubairy (2023). Do government spending multipliers depend on the sign of the shock? In *AEA Papers and Proceedings*, Volume 113, pp. 382–387. American Economic Association 2014 Broadway, Suite 305, Nashville, TN 37203.

Berger, A. N., C. Bouwman, L. Norden, R. A. Roman, G. F. Udell, and T. Wang (2024). Relationship lending and monetary policy shocks: Evidence from the u.s. Available at SSRN.

Bernanke, B. S. and A. S. Blinder (1992). The federal funds rate and the channels of monetary transmission. *The American Economic Review 82*(4), 901–921.

Bernanke, B. S. and M. Gertler (1995). Inside the black box: the credit channel of monetary policy transmission. *Journal of Economic perspectives 9*(4), 27–48.

Bianchi, F., R. Faccini, and L. Melosi (2023). A fiscal theory of persistent inflation. *The Quarterly Journal of Economics 138*(4), 2127–2179.

Bianchi, F. and L. Melosi (2019). The dire effects of the lack of monetary and fiscal coordination. *Journal of Monetary Economics 104*, 1–22.

Bolton, P., X. Freixas, L. Gambacorta, and P. E. Mistrulli (2016, October). Relationship and Transaction Lending in a Crisis. *The Review of Financial Studies 29*(10), 2643–2676.

Bonfim, D., M. A. Ferreira, F. Queiró, and S. Zhao (2025). Fiscal policy and credit supply in a crisis. *American Economic Review 115*(6), 1896–1935.

Bottero, M., S. Lenzu, and F. Mezzanotti (2020, September). Sovereign debt exposure and the bank lending channel: Impact on credit supply and the real economy. *Journal of International Economics 126*, 103328.

Brandao-Marques, L., M. Casiraghi, G. Gelos, O. Harrison, and G. Kamber (2024). Is high debt constraining monetary policy? evidence from inflation expectations. *Journal of International Money and Finance 149*, 103206.

Cantero-Saiz, M., S. Sanfilippo-Azofra, B. Torre-Olmo, and C. López-Gutiérrez (2014). Sovereign risk and the bank lending channel in europe. *Journal of International Money and Finance 47*, 1–20.

Caramp, N. and E. Feilich (2024). Monetary policy and government debt. *Journal of Money, Credit and Banking*.

Caramp, N. and D. H. Silva (2023). Fiscal policy and the monetary transmission mechanism. *Review of Economic Dynamics 51*, 716–746.

Carpenter, S. and S. Demiralp (2012). Money, reserves, and the transmission of monetary policy: Does the money multiplier exist? *Journal of macroeconomics 34*(1), 59–75.

Chen, X., E. M. Leeper, and C. Leith (2015). Us monetary and fiscal policy: Conflict or cooperation? *Manuscript, University of Glasgow, June*.

Cingano, F., F. Manaresi, and E. Sette (2016, October). Does Credit Crunch Investment Down? New Evidence on the Real Effects of the Bank-Lending Channel. *The Review of Financial Studies 29*(10), 2737–2773.

Cochrane, J. (2023). *The Fiscal Theory of the Price Level*. Princeton University Press.

Cochrane, J. H. (2022a). The fiscal roots of inflation. *Review of Economic Dynamics 45*, 22–40.

Cochrane, J. H. (2022b). A fiscal theory of monetary policy with partially-repaid long-term debt. *Review of Economic Dynamics 45*, 1–21.

Den Haan, W. J., S. W. Sumner, and G. M. Yamashiro (2007). Bank loan portfolios and the monetary transmission mechanism. *Journal of Monetary Economics 54*(3), 904–924.

Gennaioli, N., A. Martin, and S. Rossi (2014). Sovereign Default, Domestic Banks, and Financial Institutions. *The Journal of Finance 69*(2), 819–866.

Gennaioli, N., A. Martin, and S. Rossi (2018, October). Banks, government Bonds, and Default: What do the data Say? *Journal of Monetary Economics 98*, 98–113.

Goldman, J., R. Iyer, and R. Nanda (2022, December). Amplifying the Fiscal Multiplier: The Role of Banks.

Gurkaynak, R. S., B. Sack, and E. T. Swanson (2005). Do actions speak louder than words? the response of asset prices to monetary policy actions and statements. *International Journal of Central Banking 1*(1), 55–93.

Hall, G. J. and T. J. Sargent (2018). A history of us debt limits. Technical report, National Bureau of Economic Research.

Hansen, B. (2022). *Econometrics*. Princeton University Press.

Jordà, Ò. (2005). Estimation and inference of impulse responses by local projections. *American economic review 95*(1), 161–182.

Kosekova, K., A. Maddaloni, M. Papoutsi, and F. Schivardi (2025). Firm-bank relationships: a cross-country comparison. *The Review of Corporate Finance Studies*, cfaf004.

Krishnamurthy, A. and T. Muir (2017). How credit cycles across a financial crisis. Technical report, National Bureau of Economic Research.

Leeper, E. M. (1991). Equilibria under 'active'and 'passive'monetary and fiscal policies. *Journal of monetary Economics 27*(1), 129–147.

Leeper, E. M. and C. Leith (2016). Understanding inflation as a joint monetary–fiscal phenomenon. In *Handbook of macroeconomics*, Volume 2, pp. 2305–2415. Elsevier.

Lojsch, D. H., M. Rodriguez-Vives, and M. Slavik (2011). The size and composition of government debt in the euro area. Technical report, ECB Occasional Paper.

Mankiw, N. G. (2000). The savers–spenders theory of fiscal policy. *American economic review 90*(2), 120–125.

Mankiw, N. G. (2025, July). The fiscal future. Martin Feldstein Lecture, NBER Summer Institute (video). Presented July 10, 2025.

Miranda-Agrippino, S. (2016). Unsurprising shocks: information, premia, and the monetary transmission.

Morgan, D. P. (1998). The credit effects of monetary policy: Evidence using loan commitments. *Journal of Money, Credit and Banking*, 102–118.

Nakamura, E., V. Riblier, and J. Steinsson (2025). Beyond the taylor rule. Technical report, National Bureau of Economic Research.

Nakamura, E. and J. Steinsson (2014). Fiscal stimulus in a monetary union: Evidence from us regions. *American Economic Review 104*(3), 753–792.

Nakamura, E. and J. Steinsson (2018). High-frequency identification of monetary non-neutrality: the information effect. *The Quarterly Journal of Economics 133*(3), 1283–1330.

Nakamura, L. I. (1988). Customer credit, financial intermediaries and real income: preliminary evidence that credit matters. Technical report, Rutgers University Economics Working Paper.

Ramey, V. A. (2016). Macroeconomic shocks and their propagation. *Handbook of macroeconomics 2*, 71–162.

Romer, C. D. and D. H. Romer (2004). A new measure of monetary shocks: Derivation and implications. *American economic review 94*(4), 1055–1084.

Sargent, T. J. and N. Wallace (1981). Some unpleasant monetarist arithmetic. In *Federal Reserve Bank of Minneapolis Quarterly Review*, pp. 5:1–17.

Sims, C. A. (1980). Macroeconomics and reality. *Econometrica: journal of the Econometric Society*, 1–48.

Sims, C. A. (1994). A simple model for study of the determination of the price level and the interaction of monetary and fiscal policy. *Economic theory 4*(3), 381–399.

Sims, C. A. (2011). Stepping on a rake: The role of fiscal policy in the inflation of the 1970s. *European Economic Review 55*(1), 48–56.

Stock, J. H. and M. W. Watson (2010). New indexes of monthly gdp.

Wieland, J. F. and M.-J. Yang (2020). Financial dampening. *Journal of Money, Credit and Banking 52*(1), 79–113.

Woodford, M. (2003). *Interest and Prices.* Princeton University Press.

Appendix

A Data and Local Projections

A.1 Data

The data for debt is obtained at a monthly frequency from Hall and Sargent (2018). Specifically, I use the market value of total gross debt in this data to identify U.S. "debt." To turn this monthly measure into a measure of $\frac{Debt_t}{GDP_t}$, I use monthly GDP data from Stock and Watson (2010). I then normalize D_t as follows:

$$d_{t-1} = \frac{\text{Debt}_{t-1}}{\text{GDP}_{t-1}}, \quad D_{t-1} = \frac{d_{t-1} - \bar{d}}{\sigma_d}$$

This is the debt term used in the estimation of:

$$y_{t+h}^{(h)} = \beta_h \, \text{MPS}_t + \gamma_h \, D_{t-1} + \delta_h \, (\text{MPS}_t \times D_{t-1}) + \sum_{i=1}^{I} \phi_{h,i} \, \text{MPS}_{t-i} + \sum_{i=1}^{I} \boldsymbol{\theta}'_{h,i} \mathbf{X}_{t-i} + \varepsilon_{t+h} \quad \text{(A.1)}$$

A.2 Derivation of the Monetary Policy Shock Measure (Romer and Romer, 2004)

This section describes the construction of the monetary policy shock series (MPS_t) following Romer and Romer (2004). The objective is to isolate exogenous changes in monetary policy by removing systematic responses to the Federal Reserve's information set.

Let Δff_m denote the change in the Federal Reserve's intended federal funds rate around FOMC meeting m. This series is constructed from narrative and quantitative records of FOMC decisions and reflects policymakers' *intended* actions, as opposed to realized market rates. To this end, they combine data from the *Weekly Report of the Manager of Open Market Operations* with the narrative records of FOMC meetings.

Focusing on the intended changes removes high-frequency fluctuations in the observed federal funds rate unrelated to policy. It also isolates policy decisions taken at FOMC meetings. We want to eliminate the effects of what amount to largely mechanical responses to anticipated economic conditions. On this basis, the intended federal funds rate changes are regressed on the Federal Reserve's internal forecasts (known as Greenbook forecasts). The estimated equation is:

$$\begin{aligned}\Delta ff_m = \alpha + \rho\, ff_m^{\text{before}} \\ + \sum_i \beta_i \tilde{y}_{m,i} + \sum_i \gamma_i \left(\tilde{y}_{m,i} - \tilde{y}_{m-1,i}\right) \\ + \sum_i \delta_i \tilde{\pi}_{m,i} + \sum_i \theta_i \left(\tilde{\pi}_{m,i} - \tilde{\pi}_{m-1,i}\right) \\ + \phi \tilde{u}_{m,0} + \varepsilon_m,\end{aligned}$$

where $\tilde{y}_{m,i}$ is the Greenbook forecast of real output growth at horizon i, $\tilde{\pi}_{m,i}$ is the forecast of inflation, $\tilde{u}_{m,0}$ is the contemporaneous unemployment forecast, $(\tilde{y}_{m,i} - \tilde{y}_{m-1,i})$ and $(\tilde{\pi}_{m,i} - \tilde{\pi}_{m-1,i})$ are forecast revisions, and ff_m^{before} is the level of the intended funds rate prior to the meeting. The monetary policy shock is defined as the residual from the regression above:

$$\text{MPS}_m = \varepsilon_m.$$

Thus, the shock represents the component of policy that is orthogonal to the Federal Reserve's forecasts of future economic conditions. Conditioning on the Greenbook forecasts removes endogenous policy changes reacting to current or expected economic conditions, as well as anticipatory actions taken in response to expected future developments.

Under this assumption, ε_m captures exogenous variation in monetary policy. Importantly, the residual does not represent purely random noise. Instead, it reflects all influences on policy not captured by forecasts, including: shifts in policy preferences, changes in operating procedures, political or institutional considerations, and idiosyncratic decision-making factors. The estimated shocks are defined at the FOMC meeting level. For use in time-series analysis, they are converted to a monthly series by assigning each shock to the month in which the corresponding meeting occurs. If multiple meetings occur within a month, shocks are summed; if no meeting occurs, the shock is set to zero.

A.3 Bauer & Swanson 2023 Monetary Policy Shock

For robustness tests as well as tests for the post-QE period, I use the monetary policy shocks from Bauer and Swanson (2023). The data for this shock is available through the San Francisco Federal Reserve Bank's Center for Monetary Research under the heading *Monetary Policy Surprises* (MPS). Figure 1 provides a visual of the series at quarterly intervals. Positive and negative values represent expansionary and contractionary policy shocks, respectively.

The methodology from Bauer and Swanson (2023) builds upon the work of Gurkaynak et al. (2005), where MPS is identified by following changes in the prices of futures contracts in a 30-minute window around FOMC meetings. Specifically, in the method employed here, the authors follow the changes in Eurodollar futures contracts 10 minutes prior to each FOMC meeting and 20 minutes after in order to measure the "surprise" behind FOMC announcements. Consequently, this data only extends to 2023 with the end of LIBOR. However, sources exist that use a variety of alternative securities to keep this series up to date. The data begins in 1988Q1, and reaches its terminus at 2023Q4 with a total of 432 observations.

A.4 Robustness

Figure 3 gives us the differentials of the dynamics of the various macro aggregates of interest as reproduced from Caramp and Feilich (2024). That plot is constructed using the resulting IRFs from Figure 13.

One of the restrictions of the Romer and Romer (2004) data is that it is restricted to the pre-QE period. This is fine for the purposes of our study since, what follows for much of the post-QE period is a zero lower bound that would tend to obfuscate results, but it is useful to perform a robustness check using the Bauer and Swanson (2023) data. In addition to this, I'd like to use more granular banking data that controls for additional aspects of bank balance sheets and, additionally, controls for bank-level fixed effects.

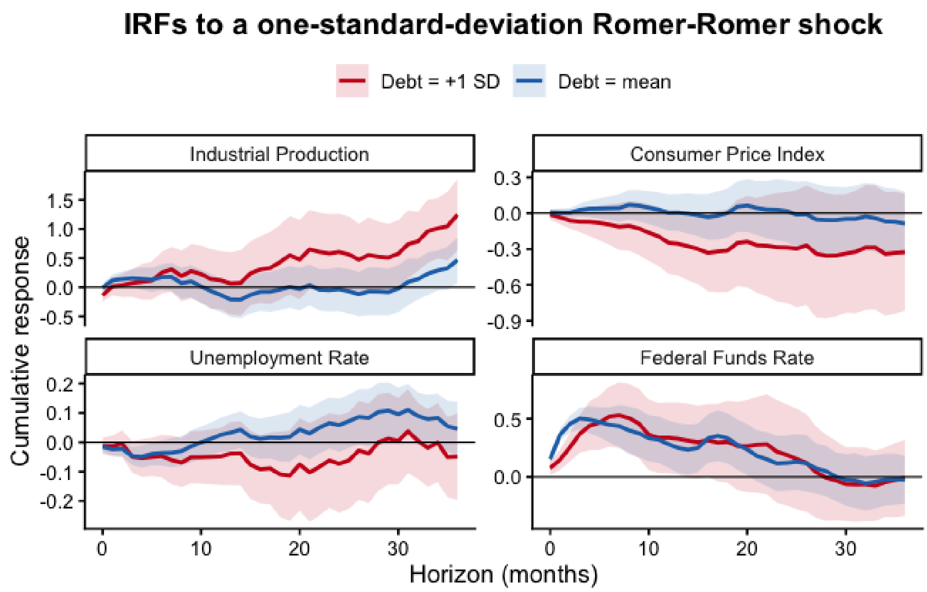


Figure 13: Macro Aggregates Response to MP Shock by Debt Regime

The data on commercial banks was gathered using the entire universe of FFIEC Call Reports lending data available through Wharton Research Data Services (WRDS) spanning 1995Q3 - 2019Q4. The data occurs at quarterly frequency and contains about 643,000 observations on about 11,000 individual units over that interval. Some of the individual units do not span the entire time series, however, as some banks went out or into business over that time. The same source allows me to also gather data on bank assets, tier 1 risk-based capital ratios, risk-weighted assets, deposits, and accumulated other comprehensive income (AOCI).

Using the WRDS bank data along with the Bauer and Swanson (2023) monetary policy shocks, I generate the results displayed in Figure 14. Note that the Bauer and Swanson (2023) shocks alter the dynamics of credit over this period, as shown in Figure 14, compared to what we observe in Figure 7; however, the story remains the same: higher debt dampens the transmission of monetary policy shocks to credit creation.

Figure 14 provides a side-by-side comparison of the state-dependent differential effects of the Bauer and Swanson (2023) shock on credit, using the WRDS data for lending and bank-level controls.

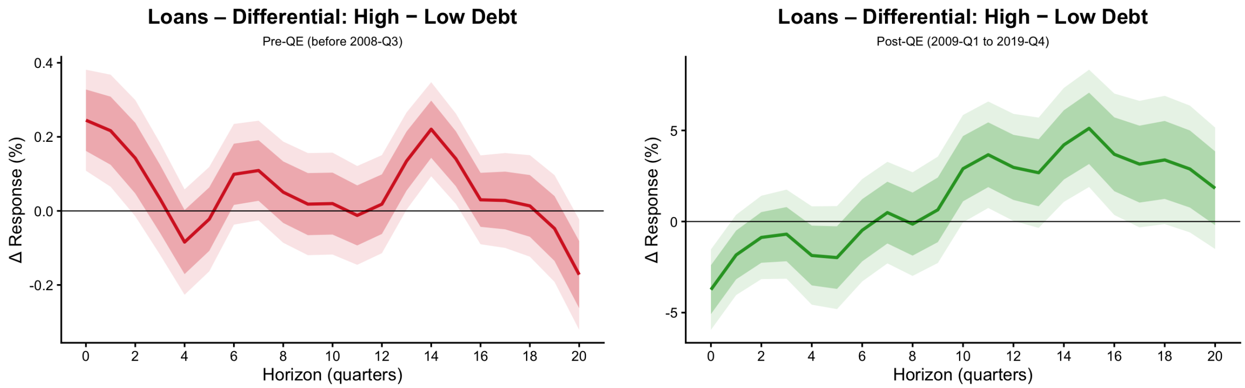


Figure 14: Pre versus Post-QE: Effects of State-Dependent MPS on Lending

I also test the effects of the Romer and Romer (2004) shock on the WRDS lending data and find similar results in Figure 15. The usual story holds: at a higher state of indebtedness, we see a muted transmission of monetary policy shocks to credit.

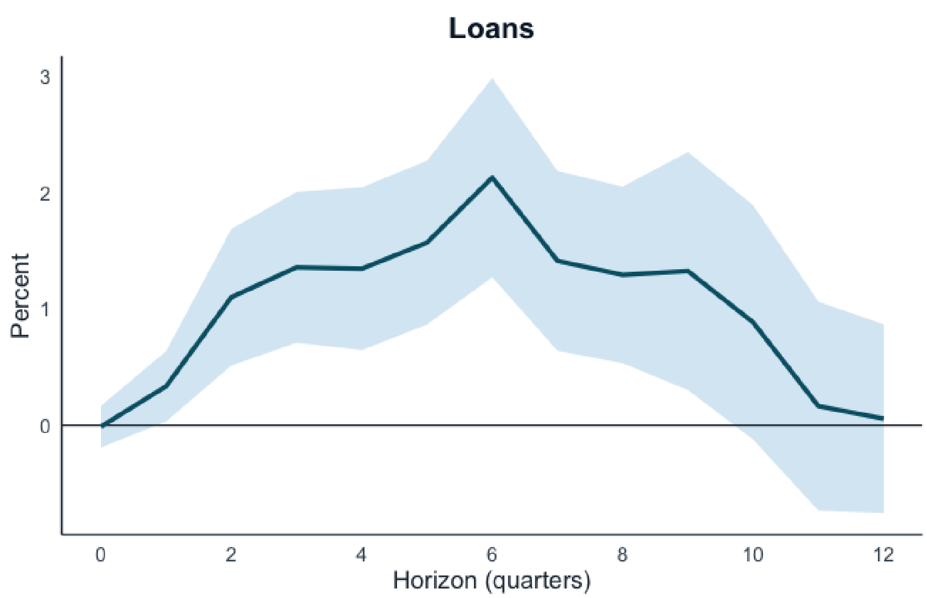


Figure 15: Pre-QE Romer Shock Differential Effects on Lending (WRDS Data)
State-dependent effects of a 1 SD monetary policy contraction of lending where the above IRF is generated by the difference between the high and low-debt states.

One thing to note, is that the reason the WRDS data was not used in the body of the paper is twofold: one, it appears at quarterly frequency, whereas the aggregate H.8 Fed data occurs at monthly frequency. Two, the WRDS data only extends back to 2001Q1. Therefore,

Figure 14 and Figure 15 lack the statistical power of the plots generated using the aggregate H.8 data that appear in the body of the paper. However, these plots are useful in that they at least do not override or cast any doubt on the main results.

As stipulated in the body of the paper, the Bayesian Information Criterion (BIC) chose a lag length of three on average over a 12-month horizon, and a lag-length of five over a 24-month horizon. I promised a plot that showed the results of choosing three lags instead of five lags below in Figure 16.

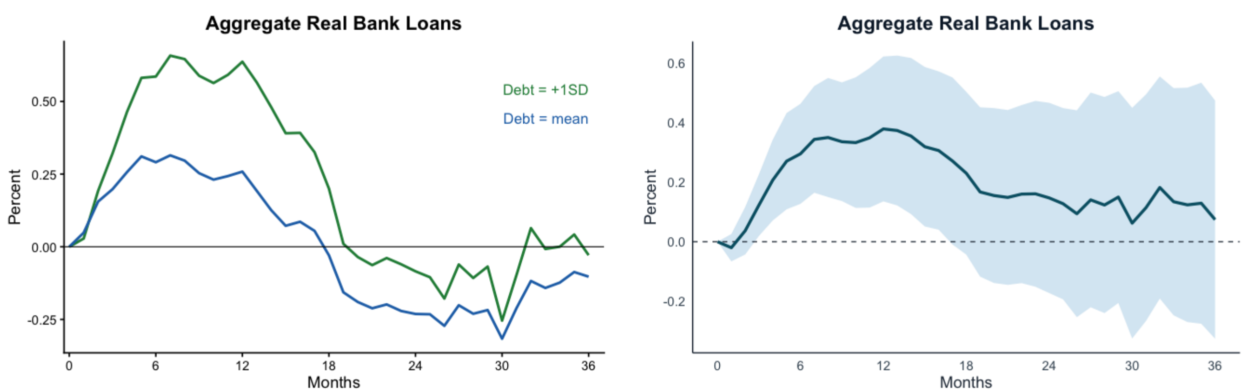


Figure 16: Three-Lag Choice for Equation 1 and Corresponding Dynamics

I also provide the impulse response of the best available proxy of the real interest rate: the Federal Reserve Bank of Cleveland's 10-Year Real Interest Rate measure. I use the standard five-lag-length specification in generating the plot:

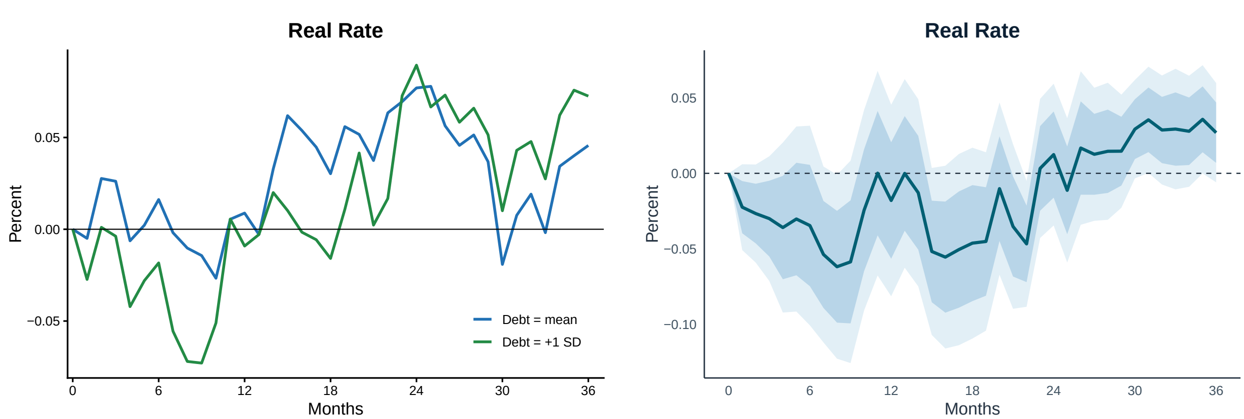


Figure 17: Real Interest Rate Response to MP Shock

B Model

B.1 Households

Household preferences are given by:

$$\max_{C_t,N_t} \mathbb{E}_t \sum_{t=0}^{\infty} \beta^t \left[\frac{C_t^{1-\sigma}}{1-\sigma} - \frac{N_t^{1+\phi}}{1+\phi} + \psi \Phi \left(\frac{D_t}{P_t} \right) \right]$$

Subject to the budget constraint:

$$P_t C_t + D_t + B_{s,t}^H + Q_t B_{\ell,t}^H = (1+i_t) B_{s,t}^H + (1+i_t^D) D_{t-1} + (1-\omega Q_t) B_{\ell,t}^H + W_t N_t + \Pi_{f,t} + \Pi_{b,t} + P_t T_t$$

Where $B_{s,t}^H$ is household holdings of short-term debt. $B_{\ell,t}^H$ are household holdings of long-term debt priced at Q_t with maturity structure governed by ω. D_t represents deposits with $\Phi'(\cdot) > 0$, $\Phi''(\cdot) < 0$.

Starting with a definition of inflation:

$$\Pi_{t+1} \equiv \frac{P_{t+1}}{P_t}$$

Goods market clearance will yield:

$$C_t = Y_t$$

From the FOC for consumption we get:

$$\lambda_t = \frac{C_t^{-\sigma}}{P_t} = \frac{Y_t^{-\sigma}}{P_t}$$

Assume Φ takes on a CRRA functional form, then we obtain the following real deposit demand condition:

$$d_t = \left(\frac{\psi}{Y_t^{-\sigma} - \beta \mathbb{E}_t \left[Y_{t+1}^{-\sigma} \frac{1+i_{t+1}^D}{\Pi_{t+1}} \right]} \right)^{\frac{1}{\nu}}$$

Where nominal deposits are defined as: $D_t = P_t d_t$

C Banks

Loan-Services Banks

Total lending is a CES aggregate of a continuum of banks:

$$L_t = \left(\int_0^1 L_t(j)^{\frac{\varepsilon-1}{\varepsilon}} dj \right)^{\frac{\varepsilon}{\varepsilon-1}} \tag{C.1}$$

That is, a competitive loan-services firm aggregates intermediate bank loans according to a CES technology.

The profit maximization problem of the loan-services firm is:

$$\max_{L_t(j)} R_t^L \left(\int_0^1 L_t(j)^{\frac{\varepsilon-1}{\varepsilon}} dj \right)^{\frac{\varepsilon}{\varepsilon-1}} - \int_0^1 R_t^L(j) L_t(j) dj \tag{C.2}$$

Taking FOCs and simplifying, we get the demand for each intermediate (i), which is proportionate to total loans, L_t:

$$L_t(j) = \left(\frac{R_t^L(j)}{R_t^L} \right)^{-\varepsilon} L_t \tag{C.3}$$

We also get an expression for the aggregate lending rate:

$$R_t^L = \left(\int_0^1 R_t^L(j)^{1-\epsilon} dj \right)^{\frac{1}{1-\varepsilon}} \tag{C.4}$$

Intermediate Banks

Intermediate banks experience nominal profits:

$$\Pi_t(j) = R_t^L(j)L_t(j) + R_{t+1}^n Q_t B_t - R_t^D(j)D_t(j) \tag{C.5}$$

subject to the capital constraint:

$$L_t(j) \leq \eta E_t(j) \tag{C.6}$$

and the budget constraint:

$$L_t(j) + Q_t B_t(j) = D_t(j) + E_t(j) \tag{C.7}$$

Plugging (53) in for $E_t(j)$ in (54) and then plugging (54) into (52), we get:

$$\Pi_t(j) = R_t^L(j)L_t(j) + [R_{t+1}^n Q_t - R_t^D(j)]B_t - \left(1 - \frac{1}{\eta}\right) R_t^D(j)L_t(j) \tag{C.8}$$

Plugging in the definition of $L_t(j)$ from (50) into (55), we get:

$$\Pi_{t+1}(j) = R_t^L(j)\left(\frac{R_t^L(j)}{R_t^L}\right)^{-\varepsilon} L_t + [R_{t+1}^n Q_t - R_t^D(j)]B_t - \left(1 - \frac{1}{\eta}\right) R_t^D(j)\left(\frac{R_t^L(j)}{R_t^L}\right)^{-\varepsilon} L_t \tag{C.9}$$

Which simplifies to:

$$\Pi_t(j) = \left(R_t^L(j)^{1-\varepsilon} R_t^{L\,\varepsilon} - \left(1 - \frac{1}{\eta}\right) R_t^D(j)R_t^L(j)^{-\varepsilon} R_t^{L\,\varepsilon}\right) L_t + [R_{t+1}^n Q_t - R_t^D(j)]B_t \tag{C.10}$$

Taking the derivative with respect to $R_t^L(j)$:

$$\frac{\partial \Pi_t(j)}{\partial R_t^L(j)} = L_t R_t^{L\,\varepsilon}\left[(1-\varepsilon)R_t^L(j)^{-\varepsilon} + \varepsilon\left(1 - \frac{1}{\eta}\right) R_t^D(j)R_t^L(j)^{-\varepsilon-1}\right]. \tag{C.11}$$

Setting the FOC equal to zero:

$$(1-\varepsilon)R_t^L(j)^{-\varepsilon} + \varepsilon\left(1-\frac{1}{\eta}\right)R_t^D(j)R_t^L(j)^{-\varepsilon-1} = 0. \tag{C.12}$$

Multiply both sides by $R_t^L(j)^{\varepsilon+1}$:

$$(1-\varepsilon)R_t^L(j) + \varepsilon\left(1-\frac{1}{\eta}\right)R_t^D(j) = 0. \tag{C.13}$$

Therefore, the optimal lending rate is:

$$R_t^L(j) = \underbrace{\frac{\varepsilon}{\varepsilon-1}}_{\text{markup}}\left(1-\frac{1}{\eta}\right)R_t^D(j). \tag{C.14}$$

Deposits

Similarly, savings products sold by banks and form an aggregate bank deposit D_t:

$$D_t = \left(\int_0^1 D_t(j)^{\frac{\varepsilon^D-1}{\varepsilon^D}}\right)^{\frac{\varepsilon^D}{\varepsilon^D-1}} \tag{C.15}$$

Then the problem is similar to the loan problem above, except, for loans you subtract payments to banks, for deposits you add receipts from banks:

$$\max_{\{D_t(j)\}} \quad \int_0^1 R_t^D(j)\, D_t(j)\, dj \;-\; R_t^D\left(\int_0^1 D_t(j)^{\frac{\varepsilon_D-1}{\varepsilon_D}}\, dj\right)^{\frac{\varepsilon_D}{\varepsilon_D-1}} \tag{C.16}$$

Similar to the process above of the firm that provides loan services, we end up with an expression for deposits in the form of:

$$D_t(j) = \left(\frac{R_t^D(j)}{R_t^D}\right)^{\varepsilon^D} D_t \tag{C.17}$$

$$\tag{C.18}$$

Now, using (54) to eliminate $Q_t B_t(j)$ in (52), we get:

$$\Pi_{t+1}(j) = R_t^L(j)L_t(j) + R_{t+1}^n(D_t(j) + E_t(j) - L_t(j)) - R_t^D(j)D_t(j) \tag{C.19}$$

Collecting terms:

$$\Pi_{t+1}(j) = [R_t^L(j) - R_{t+1}^n]L_t(j) + [R_{t+1}^n - R_t^D(j)]D_t(j) + R_{t+1}^n E_t(j) \tag{C.20}$$

Plugging in (64) into (67):

$$\Pi_{t+1}(j) = [R_t^L(j) - R_{t+1}^n]L_t(j) + [R_{t+1}^n - R_t^D(j)]\left(\frac{R_t^D(j)}{R_t^D}\right)^{\varepsilon^D} D_t + R_{t+1}^n E_t(j) \tag{C.21}$$

Simplifying:

$$\Pi_{t+1}(j) = [R_t^L(j) - R_{t+1}^n]L_t(j) + R_{t+1}^n R_t^D(j)^{\varepsilon^D} R_t^{D\ -\varepsilon^D} D_t - R_t^D(j)^{1+\varepsilon^D} R_t^{D\ -\varepsilon^D} D_t + R_{t+1}^n E_t(j) \tag{C.22}$$

Taking the derivative with respect to $R_t^D(j)$:

$$\frac{\partial \Pi_{t+1}(j)}{\partial R_t^D(j)} = \varepsilon_D R_{t+1}^n R_t^D(j)^{\varepsilon_D - 1} R_t^{D\,-\varepsilon_D} D_t - (1+\varepsilon_D) R_t^D(j)^{\varepsilon_D} R_t^{D\,-\varepsilon_D} D_t \tag{C.23}$$

Factor out common terms:

$$\frac{\partial \Pi_{t+1}(j)}{\partial R_t^D(j)} = R_t^{D\,-\varepsilon_D} D_t \left[\varepsilon_D R_{t+1}^n R_t^D(j)^{\varepsilon_D - 1} - (1+\varepsilon_D) R_t^D(j)^{\varepsilon_D}\right] \tag{C.24}$$

Setting the FOC equal to zero:

$$\varepsilon_D R_{t+1}^n R_t^D(j)^{\varepsilon_D - 1} - (1+\varepsilon_D) R_t^D(j)^{\varepsilon_D} = 0 \tag{C.25}$$

Factor $R_t^D(j)^{\varepsilon_D - 1}$:

$$R_t^D(j)^{\varepsilon_D - 1}\left[\varepsilon_D R_{t+1}^n - (1+\varepsilon_D) R_t^D(j)\right] = 0 \tag{C.26}$$

Solving for the optimal deposit rate:

$$R_t^D(j) = \frac{\varepsilon_D}{1+\varepsilon_D} R_{t+1}^n \tag{C.27}$$

If $E_t[R_{t+1}^n] = 1 + i_t$, then

$$R_t^D(j) = \frac{\varepsilon_D}{\varepsilon_D + 1}(1 + i_t) \tag{C.28}$$

Which does not depend on j. Therefore, our optimal deposit rate is:

$$R_t^{D^*} = \frac{\varepsilon_D}{\varepsilon_D + 1}(1 + i_t) \tag{C.29}$$

D Calibration of Parameters

D.1 Moments:

Table 4: Key Empirical Moments Used to Map Data Moments into Model Shocks

Moment	Value	
Mean debt-to-GDP, $\bar{d}$	0.5109	ratio
Standard deviation of debt-to-GDP, σ_d	0.1452	ratio
High-debt state, $\bar{d} + \sigma_d$	0.6561	ratio
Debt increase relative to mean, $\sigma_d/\bar{d}$	0.2842	percent of mean
Mean Romer–Romer shock, $\overline{\varepsilon^{MP}}$	-0.0008	percent
Standard deviation of Romer–Romer shock, $\sigma_{\varepsilon^{MP}}$	0.3017	percent
VAR monetary policy shock	0.5128	percent

Notes: Sample period is 1973:02–2007:12. Debt moments are computed from the debt-to-GDP series used to define the high-debt regime. Romer–Romer moments are computed from the monthly `resid_romer` series over the same sample. The model calibration uses the proportional debt deviation, 0.2842, and the one-standard-deviation monetary policy shock in decimal form, 0.0030.

D.2 Basic Parameters:

Table 5 reports the baseline calibration *before taking the model to the data and matching moments*. The discount factor $\beta = 0.99$ corresponds to a standard quarterly frequency. The relative risk aversion $\sigma = 3$ and inverse Frisch elasticity $\phi = 1$ are within the range commonly used in the New Keynesian literature. The slope of the Phillips curve $\kappa = 0.5$ implies moderate price stickiness. Bank leverage $\eta = 10$ is consistent with typical pre-crisis US commercial bank balance sheets. The loan-market elasticity $\varepsilon = 7$ and deposit-market elasticity $\varepsilon_D = 1$ are calibrated to match a US loan-deposit spread of approximately 300 basis points. The weight on loan-rate pass-through $\xi = 0.75$ is tentative and subject to revision.[11]

[11] $\xi \equiv \bar{i}^L/(\bar{i}^L + \delta)$ is pinned down by the steady-state loan rate and depreciation rate; the value reported here is an approximation pending full steady-state calibration.

Table 5: Basic Parameter Calibration

Parameter	Value	Parameter	Value
Households & Firms			
β	0.99	α	0.33
σ	1.5	ψ	1
ϕ	1	κ	0.50
Banks			
ε	7	ε_D	1
η	10	ξ	$0.75^{\dagger}$
Monetary & Fiscal Policy Rules			
$\theta_{i\pi}$	0.80	$\theta_{s\pi}$	0.25
θ_{ix}	0.50	θ_{sx}	1.00
Asset Pricing			
ω	0.70	α_v	0.20
γ	0.80		
Shock Processes			
ρ_a	0.90	ρ^s	0.40
ρ^i	0.70		

†Tentative.

D.3 Composite Parameters

The composite parameters reported in Table 6 are computed analytically from the basic calibration above. Θ_L summarizes the net markup applied to the policy rate in the loan rate equation (100); Λ_π rescales the forward inflation term in the New Keynesian Phillips Curve once loan-financed capital is accounted for; and Γ_π, Ω_y, Ω_a, Γ_u are the reduced-form coefficients on lagged inflation, output, productivity, and the monetary shock, respectively, in the Phillips Curve (111).

E Solving the Model

E.1 Combining Conditions

The purpose of this section is to solve the model laid out in the previous section. The model can be expressed in the following form:

Table 6: Composite Parameters

Parameter	Value
Θ_L	0.525
Λ_π	1.185
Γ_π	1.019
Ω_y	-1.023
Ω_a	1.008
Γ_u	0.099

$$Ay_{t+1} = By_t + C\varepsilon_{t+1} + D\delta_{t+1}$$

Start with the IS equation:

$$y_{t+1} + \sigma\pi_{t+1} = y_t + \sigma(\theta_{i\pi}\pi_t + \theta_{ix}y_t + u_t^i) + \delta_{t+1}^y + \sigma\delta_{t+1}^\pi \tag{E.1}$$

Then the NKPC:

$$\pi_{t+1} = \frac{1}{\beta}\pi_t - \frac{\kappa}{\beta}mc_t + \delta_{t+1}^\pi$$

Which can be decomposed as follows:

$$\pi_{t+1} = \frac{1}{\beta}\pi_t - \frac{\kappa}{\beta}[(1-\alpha)(\sigma y_t + \phi n_t) + \alpha\xi\, r_{t+1}^L - a_t] + \delta_{t+1}^\pi$$

From equation (90), call $\frac{\varepsilon}{\varepsilon-1}\left(1-\frac{1}{\eta}\right)\frac{\varepsilon_D}{\varepsilon_D+1} \equiv \Theta_L$. Then we have:

$$\pi_{t+1} = \frac{1}{\beta}\pi_t - \frac{\kappa}{\beta}\left[(1-\alpha)\left(\left(\sigma + \frac{\phi}{1-\alpha}\right)y_t - \frac{\phi\alpha}{1-\alpha}k_t\right) + \alpha\xi\left(\Theta_L i_t - \pi_{t+1}\right) - \left(1+\frac{\phi}{1-\alpha}\right)a_t\right] + \delta_{t+1}^\pi$$

Then this expands to:

$$\pi_{t+1} = \frac{1}{1+\alpha\xi}\left\{\left(\frac{1+\kappa(1-\alpha)\alpha\xi\Theta_L\theta_{i\pi}}{\beta}\right)\pi_t - \frac{\kappa}{\beta}\left[(1-\alpha)\left(\left(\sigma+\frac{\phi}{1-\alpha}+\alpha\xi\Theta_L\theta_{ix}\right)y_t - \frac{\phi\alpha}{1-\alpha}k_t\right)\right.\right.$$

$$+\alpha\xi\,\Theta_L u_t^i - \left(1+\frac{\phi}{1-\alpha}\right)a_t\Bigg] + \delta_{t+1}^{\pi}\} \quad \text{(E.2)}$$

Then our equation for k_t:

$$\begin{aligned}
k_t &= n_t + w_t - r_t^K \\
&= n_t + \sigma y_t + \phi n_t - r_t^K \\
&= \frac{1+\phi}{1-\alpha}\left(y - a_t - \alpha k_t\right) + \sigma y_t - r_t^K \\
&= \left(\frac{1+\phi}{1-\alpha} + \sigma\right) y_t - \frac{1+\phi}{1-\alpha} a_t - \frac{\alpha(1+\phi)}{1-\alpha} k_t - r_t^K \\
\left(1+\frac{\alpha(1+\phi)}{1-\alpha}\right) k_t &= \left(\frac{1+\phi}{1-\alpha} + \sigma\right) y_t - \frac{1+\phi}{1-\alpha} a_t - \xi[\Theta_L(\theta_{i\pi}\pi_t + \theta_{ix} y_t + u_t^i) - \pi_{t+1}]
\end{aligned}$$

Which yields our expanded condition for k_t:

$$\left(1+\frac{\alpha(1+\phi)}{1-\alpha}\right) k_t + \xi\pi_{t+1} = \left(\frac{1+\phi}{1-\alpha} + \sigma + \xi\Theta_L\theta_{ix}\right) y_t - \frac{1+\phi}{1-\alpha} a_t - \xi[\Theta_L(\theta_{i\pi}\pi_t + u_t^i)]$$

Rearranging:

$$k_t = \frac{1}{\left(1+\frac{\alpha(1+\phi)}{1-\alpha}\right)}\left[\left(\frac{1+\phi}{1-\alpha} + \sigma + \xi\Theta_L\theta_{ix}\right) y_t - \frac{1+\phi}{1-\alpha} a_t - \xi[\Theta_L(\theta_{i\pi}\pi_t + u_t^i) - \pi_{t+1}]\right] \quad \text{(E.3)}$$

We can plug this in for k_t in equation (99), which gives us:

$$\begin{aligned}
\pi_{t+1} &= \frac{1}{1+\alpha\xi}\Bigg\{\left(\frac{1+\kappa(1-\alpha)\alpha\xi\Theta_L\theta_{i\pi}}{\beta}\right)\pi_t \\
&-\frac{\kappa}{\beta}\Bigg[(1-\alpha)\Bigg(\left(\sigma + \frac{\phi}{1-\alpha} + \alpha\xi\Theta_L\theta_{ix}\right) y_t - \frac{\phi\alpha}{1-\alpha}\frac{1}{\left(1+\frac{\alpha(1+\phi)}{1-\alpha}\right)}\left[\left(\frac{1+\phi}{1-\alpha} + \sigma + \xi\Theta_L\theta_{ix}\right) y_t - \frac{1+\phi}{1-\alpha} a_t - \xi[\Theta_L(\theta_{i\pi}\pi_t + u_t^i) - \pi_{t+1}]\right]\Bigg) \\
&+\alpha\xi\,\Theta_L u_t^i - \left(1+\frac{\phi}{1-\alpha}\right)a_t\Bigg] + \delta_{t+1}^{\pi}\Bigg\}
\end{aligned}$$

Collecting terms we have:

$$\left[(1+\alpha\xi) - \frac{\kappa\phi\alpha\xi}{\beta\left(1+\frac{\alpha(1+\phi)}{1-\alpha}\right)}\right]\pi_{t+1} = \left[\frac{1+\kappa(1-\alpha)\alpha\xi\Theta_L\theta_{i\pi}}{\beta} - \frac{\kappa\phi\alpha\xi\Theta_L\theta_{i\pi}}{\beta\left(1+\frac{\alpha(1+\phi)}{1-\alpha}\right)}\right]\pi_t$$

$$+\left[-\frac{\kappa}{\beta}(1-\alpha)\left(\sigma+\frac{\phi}{1-\alpha}+\alpha\xi\Theta_L\theta_{ix}\right)+\frac{\kappa\phi\alpha}{\beta\left(1+\frac{\alpha(1+\phi)}{1-\alpha}\right)}\left(\frac{1+\phi}{1-\alpha}+\sigma+\xi\Theta_L\theta_{ix}\right)\right]y_t$$

$$+\left[-\frac{\kappa\phi\alpha}{\beta\left(1+\frac{\alpha(1+\phi)}{1-\alpha}\right)}\left(\frac{1+\phi}{1-\alpha}\right)+\frac{\kappa}{\beta}\left(1+\frac{\phi}{1-\alpha}\right)\right]a_t$$

$$-\left[\frac{\kappa\phi\alpha\xi\Theta_L}{\beta\left(1+\frac{\alpha(1+\phi)}{1-\alpha}\right)}+\frac{\kappa\alpha\xi\Theta_L}{\beta}\right]u_t^i+\frac{1}{1+\alpha\xi}\delta_{t+1}^{\pi}$$

Consolidating the parameter blocks into single parameters:

$$\Lambda_\pi \equiv \left[(1+\alpha\xi)-\frac{\kappa\phi\alpha\xi}{\beta\left(1+\frac{\alpha(1+\phi)}{1-\alpha}\right)}\right]$$

$$\Gamma_\pi \equiv \left[\frac{1+\kappa(1-\alpha)\alpha\xi\Theta_L\theta_{i\pi}}{\beta}-\frac{\kappa\phi\alpha\xi\Theta_L\theta_{i\pi}}{\beta\left(1+\frac{\alpha(1+\phi)}{1-\alpha}\right)}\right]$$

$$\Omega_y \equiv \left[-\frac{\kappa}{\beta}(1-\alpha)\left(\sigma+\frac{\phi}{1-\alpha}+\alpha\xi\Theta_L\theta_{ix}\right)+\frac{\kappa\phi\alpha}{\beta\left(1+\frac{\alpha(1+\phi)}{1-\alpha}\right)}\left(\frac{1+\phi}{1-\alpha}+\sigma+\xi\Theta_L\theta_{ix}\right)\right]$$

$$\Omega_a \equiv \left[-\frac{\kappa\phi\alpha}{\beta\left(1+\frac{\alpha(1+\phi)}{1-\alpha}\right)}\left(\frac{1+\phi}{1-\alpha}\right)+\frac{\kappa}{\beta}\left(1+\frac{\phi}{1-\alpha}\right)\right]$$

$$\Gamma_u \equiv \left[\frac{\kappa\phi\alpha\xi\Theta_L}{\beta\left(1+\frac{\alpha(1+\phi)}{1-\alpha}\right)}+\frac{\kappa\alpha\xi\Theta_L}{\beta}\right]$$

We end up with a simpler equation for π_{t+1}:

$$\Lambda_\pi\pi_{t+1}=\Gamma_\pi\pi_t+\Omega_y y_t+\Omega_a a_t-\Gamma_u u_t^i+\frac{1}{1+\alpha\xi}\delta_{t+1}^{\pi} \tag{E.4}$$

From (92) and (93) we get our expression for bonds:

$$\omega q_{t+1}=\theta_{ix}y_t+\theta_{i\pi}\pi_t+q_t+u_t^i+\omega\delta_{t+1}^{q} \tag{E.5}$$

From equation (94)-(96) and (101) we get:

$$\theta_{sx}y_{t+1}+\theta_{s\pi}\pi_{t+1}+\left(1+\frac{\alpha_v}{\gamma}\right)v_{t+1}^{*}+u_{t+1}^{s}=$$

$$\left(\theta_{i\pi} - \frac{\Gamma_\pi}{\Lambda_\pi}\right)\pi_t + \left(\theta_{ix} - \frac{\Omega_y}{\Lambda_\pi}\right)y_t + \left(1 + \frac{\Gamma_u}{\Lambda_\pi}\right)u_t^i - \Omega_a a_t + v_t^* \quad \text{(E.6)}$$

Then we also get, from equation (96):

$$\theta_{sx}y_{t+1} + (1+\theta_{s\pi})\pi_{t+1} + \alpha v_{t+1}^* + v_{t+1} - \omega q_{t+1} + u_{t+1}^s = v_t - q_t \quad \text{(E.7)}$$

We also have our evolution of TFP:

$$a_{t+1} = \rho_a a_t + \epsilon_{t+1}^\alpha \quad \text{(E.8)}$$

Equations (98) - (104) are, therefore, the equations that will comprise our system.

Now we can proceed with solving the model as in the matrix notation above.

E.2 Determinacy Condition

The model is written as

$$\underbrace{\mathbf{A}}_{6\times6}\mathbf{z}_{t+1} = \underbrace{\mathbf{B}}_{6\times6}\mathbf{z}_t + \underbrace{\mathbf{C}}_{6\times3}\boldsymbol{\varepsilon}_{t+1} + \underbrace{\mathbf{D}}_{6\times3}\boldsymbol{\eta}_{t+1}.$$

Therefore, we have:

$$\begin{bmatrix} 1 & \sigma & 0 & 0 & 0 & 0 \\ 0 & \Lambda_\pi & 0 & 0 & 0 & 0 \\ 0 & 0 & \omega & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} y_{t+1} \\ \pi_{t+1} \\ q_{t+1} \\ a_{t+1} \\ u_{t+1}^i \\ u_{t+1}^s \end{bmatrix}$$

$$
= \begin{bmatrix} 1+\sigma\theta_{ix} & \sigma\theta_{i\pi} & 0 & 0 & \sigma & 0 \\ \Omega_y & \Gamma_\pi & 0 & \Omega_a & -\Gamma_u & 0 \\ \theta_{ix} & \theta_{i\pi} & 1 & 0 & 1 & 0 \\ 0 & 0 & 0 & \rho_a & 0 & 0 \\ 0 & 0 & 0 & 0 & \rho_i & 0 \\ 0 & 0 & 0 & 0 & 0 & \rho_s \end{bmatrix} \begin{bmatrix} y_t \\ \pi_t \\ q_t \\ a_t \\ u_t^i \\ u_t^s \end{bmatrix}
$$

$$
+ \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} \varepsilon_{t+1}^a \\ \varepsilon_{t+1}^i \\ \varepsilon_{t+1}^s \end{bmatrix} + \begin{bmatrix} 1 & 0 \\ 0 & 0 \\ 0 & \omega \\ 0 & 0 \\ 0 & 0 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} \delta_{t+1}^y \\ \delta_{t+1}^q \end{bmatrix}
$$

Inverting the A matrix and moving to the RHS, we end up with:

$$
z_{t+1} = A^{-1}B\, z_t + A^{-1}C\, \varepsilon_{t+1} + A^{-1}D\, \eta_{t+1}
$$

Where:

$$
A^{-1}B = \begin{bmatrix} \frac{\Lambda_\pi - \Omega_y\sigma + \Lambda_\pi\sigma\theta_{ix}}{\Lambda_\pi} & -\frac{\sigma(\Gamma_\pi - \Lambda_\pi\theta_{i\pi})}{\Lambda_\pi} & 0 & -\frac{\Omega_a\sigma}{\Lambda_\pi} & \frac{\sigma(\Gamma_u + \Lambda_\pi)}{\Lambda_\pi} & 0 \\ \frac{\Omega_y}{\Lambda_\pi} & \frac{\Gamma_\pi}{\Lambda_\pi} & 0 & \frac{\Omega_a}{\Lambda_\pi} & -\frac{\Gamma_u}{\Lambda_\pi} & 0 \\ \frac{\theta_{ix}}{\omega} & \frac{\theta_{i\pi}}{\omega} & \frac{1}{\omega} & 0 & \frac{1}{\omega} & 0 \\ 0 & 0 & 0 & \rho_a & 0 & 0 \\ 0 & 0 & 0 & 0 & \rho_i & 0 \\ 0 & 0 & 0 & 0 & 0 & \rho_s \end{bmatrix}
$$

The eigenvalues of $A^{-1}B$ are as follows: 1.1111, 0.8250, 3.0799, 0.90, 0.70, 0.50, which confirms that we have two explosive roots for two forward looking variables.

It is instructive to examine the determinacy condition a little more closely. To find the conditions for determinacy, I follow the procedure laid out in Woodford (2001). First, let's say $A^{-1}B \equiv \Theta$. Then, an eigenvalue satisfies:

$$\det(\mu I - \Theta) = 0$$

The factorization of the characteristic polynomial yields:

$$\det(\mu I - \Theta) = \frac{(\mu\omega - 1)(\mu - \rho_a)(\mu - \rho_i)(\mu - \rho_s)}{\Lambda_\pi\,\omega} \cdot \Big[\Gamma_\pi - \Gamma_\pi\mu - \Lambda_\pi\mu + \Lambda_\pi\mu^2 + \Gamma_\pi\sigma\theta_{ix} + \Omega_y\mu\sigma - \Omega_y\sigma\theta_{i\pi} - \Lambda_\pi\mu\sigma\theta_{ix}\Big]$$

Evaluating the characteristic polynomial at $\mu = 1$, we get:

$$p(1) \equiv \det(I - \Theta) = -\frac{\sigma(\omega - 1)(\rho_a - 1)(\rho_i - 1)(\rho_s - 1)}{\Lambda_\pi\,\omega} \cdot \Big[\Omega_y + \Gamma_\pi\theta_{ix} - \Lambda_\pi\theta_{ix} - \Omega_y\theta_{i\pi}\Big]$$

Given our calibration, $-\frac{\sigma(\omega-1)(\rho_a-1)(\rho_i-1)(\rho_s-1)}{\Lambda_\pi\,\omega}$ turns out to be negative as a whole. We have two forward looking variables and we know that the NKPC gives us an eigenvalue greater than 1. Therefore, in order to obtain determinacy, we must have $p(1) > 0$. Hence, the determinacy condition becomes:

$$\boxed{\Omega_y(1 - \theta_{i\pi}) + \theta_{ix}(\Gamma_\pi - \Lambda_\pi) < 0}$$

Which, given our calibrations, becomes:

$$-1.023(1 - 0.8) + 0.5(1.019 - 1.185) = -0.2876 < 0$$

Satisfying the determinacy condition.

E.3 Additional Analysis

Below, in Figure 18, we see that, in response to an exogenous one percentage point increase in interest rates, lending contracts less and less as future surpluses fall ever further from trend and, therefore, debt-to-GDP rises above trend. Therefore, lending becomes less sensitive to a given monetary policy contraction as indebtedness rises.

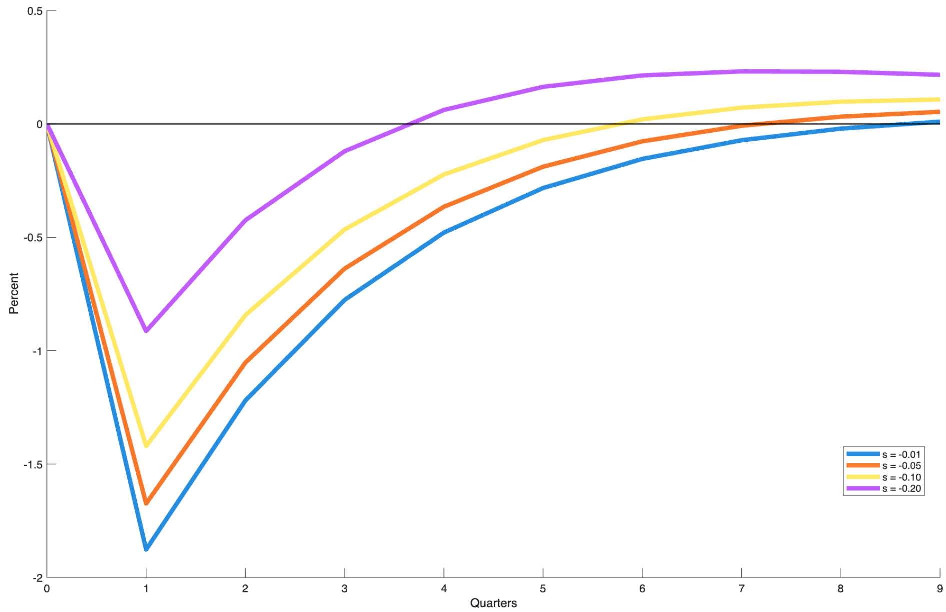


Figure 18: Credit Response to a 1pp MP Shock at Varying Debt/GDP Levels

Contractionary MP shock's effect on credit at four different deviations of debt from trend. The variable "s" represents a decline in future surpluses of the following magnitudes: 1 percent, 5 percent, 10 percent, and 20 percent.

The reason for this is fairly straightforward. A surplus shock generates inflation that, while partly offset by fluctuations in bond prices, spurs economic activity. The sign of inflation, as debt rises, continues to trend in the positive direction, despite the monetary contraction. As debt-to-GDP deviates far enough from trend, real interest rates rise less and less for a given monetary policy shock.

In the model, by the time expected future surpluses fall 10% below trend, inflation actually begins to increase in the face of a 1 percent monetary policy contraction. As monetary policy generates an ever smaller impact on real interest rates ($r = i - \pi$), lending likewise responds by less. This can be observed in the two panels of Figure 19.

As indebtedness rises, it creates additional inflationary pressure, which pushes down real interest rates, thus offsetting the effects of the monetary contraction. Specifically, for every 5% deviation of surpluses below trend, we see real interest rates r rise by 0.15% less after the 1% monetary policy shock. In the extreme, the monetary policy shock only slightly dampens what is otherwise a large inflationary spike in the wake of the rise in debt; reaching 3% as surpluses fall 20% below trend, even though nominal interest rates are held 1% above trend. Therefore, in order to achieve the same desired fall in economic activity, monetary policy will have to become more contractionary as debt-to-GDP rises.

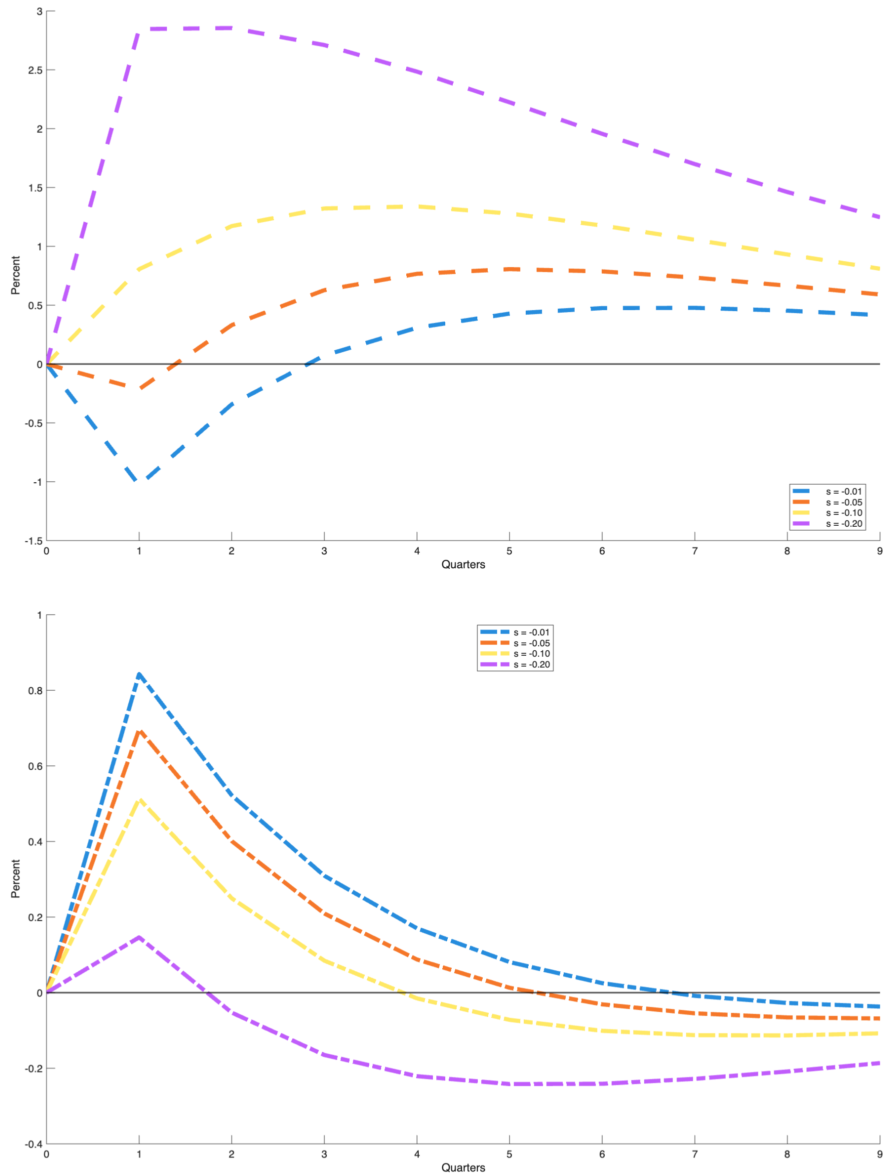


Figure 19: State-Dependent IRFs of Inflation π and Real Interest Rates r

Top panel represents contractionary MP shock's effects on inflation π and bottom panel represents the same shock's effects on real interest rates r at four different deviations of debt from trend. Real interest rates are $r = i - \pi$, with i as the policy rate.

At first glance, it appears that the solution to this problem would be for the monetary authority to simply react more strongly to inflation as debt rises and that this will resolve the problem. In reality, although this approach will generate a larger contraction in the immediate term, it will generate very substantial inflation over the long run. The reason is that the value of debt increases in interest rates. As the monetary authority pushes up rates, the government's borrowing costs also rise. Therefore, although responding more strongly to inflation by pushing interest rates even higher to meet a given inflation target will generate the desired result in the near term, this will result in substantial and drawn-out inflation over the medium and long term.

Figure 20 illustrates precisely what the problem is and, conveniently, helps shed light on why my results, along with those of Caramp and Feilich (2024) differ from studies of the European experience Cantero-Saiz et al. (2014). Notice the consequences of ramping up the policy rate as debt grows. As the monetary policy contraction gets larger jointly with debt-to-GDP, both lending and inflation contract by more over near-term horizons. At the extreme end — 20% increase in policy rates jointly with a 20% decline in future surpluses — we see as much as a 37.5% drop in total lending and 20.5% fall in inflation. Clearly this is extreme, but it illustrates the point: if monetary policy contracts more strongly, in step with rising indebtedness, it can generate strong near-term contractions in both inflation and lending.

However, and this is the key point, this policy prescription comes at the expense of massive and persistent inflation as soon as one year into the future. Observe the dashed lines. Over the immediate term, within a year, inflation falls below trend. However, since the freshly accumulated debt is only partially repaid with future surpluses, our valuation equation (8) dictates that bond prices and inflation have to absorb the residual. Although long-term debt acts as a shock absorber Cochrane (2022a); Leeper and Leith (2016), the joint effects of a contractionary monetary policy shock and negative surplus shock is to generate larger inflation in the future.

The mechanism is familiar. In an environment of fiscal dominance where the government issues nominal debt,[12] Sims (1994); Leeper and Leith (2016) explain how monetary and fiscal policy jointly determine the price level. Higher interest rates generate higher debt service costs for the government. If the fiscal authority does not adjust surpluses appropriately, then

[12] Fiscal dominance — Active Fiscal/Passive Monetary Policy

in order to ensure government solvency, the price level adjusts so that the valuation equation (8) is satisfied. This works through a wealth effect channel. If the government has to offer increasingly higher interest to bond holders, barring a future tax increase to finance these interest payments, households (bond holders) receive a windfall in income that translates into additional demand and, thus, prices. Hence the sense in which people understand FTPL to be "non-Ricardian."

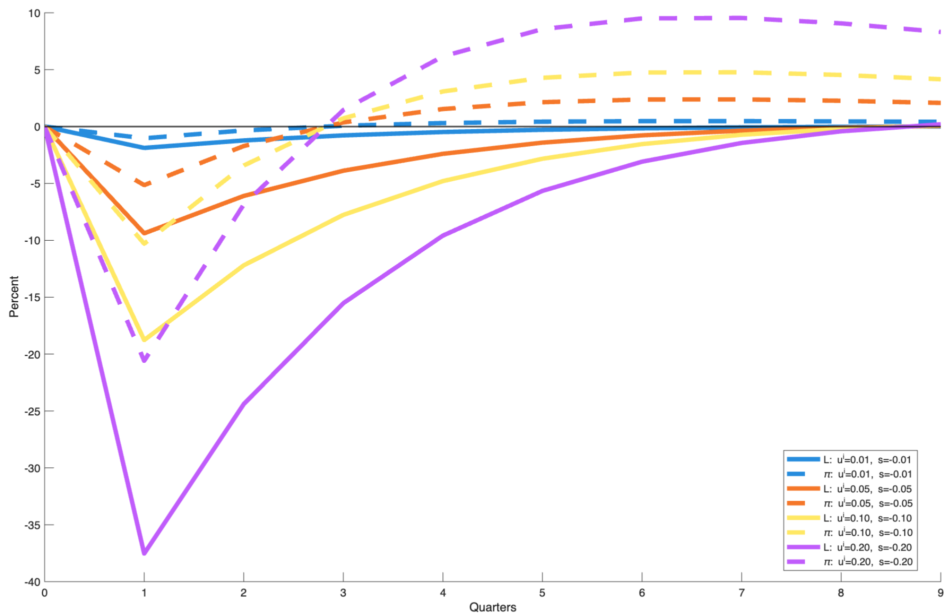


Figure 20: Lock-Step MP + FP Shocks

Dashed lines represent inflation π. Solid lines represent lending L. Each line is the response of a given variable to a joint contractionary monetary policy shock and negative fiscal shock (fall in surpluses) of the same magnitude.

In addition to the wealth effects of an active fiscal regime, inflation expectations also play a key role in determining the path of realized inflation. Chen et al. (2015) explain that inflation expectations are anchored only partially by monetary policy — they are also anchored by expected fiscal backing and regime credibility. They work with a model where firm marginal costs depend positively on tax rates. If people expect the government will raise distortionary taxes in the future, they expect firms' costs to rise, so they expect inflation. That expectation feeds into current inflation. To counteract that, the fiscal authority cuts taxes today, but doing so increases debt, which makes future tax hikes even more likely, creating a destabilizing loop.

These mechanisms, taken together, explain the dynamics of credit in Figure 18 and Figure 20. In the former case, monetary policy responds highly passively to each successive increase in debt-to-GDP. That is, we observe how the dynamics of credit evolve over time when, at each successively higher level of indebtedness, the monetary authority's response to inflation at any given state is the same (1% interest rate hike). In the latter case, as monetary policy responds more aggressively to each successive deterioration in fiscal backing, inflation falls by more in the immediate horizon, pushing down credit accumulation, but at the expense of greater future inflation.

These two cases help cast light on the differences between the U.S. and European experiences in terms of how monetary and fiscal policy jointly determine credit dynamics. In the run up to the sovereign debt crisis of the 2010s, high sovereign-risk countries saw credit contract more strongly to a given monetary policy shock than low sovereign-risk countries. This is related to the deterioration of bank balance sheets in those countries who held substantial quantities of these high-risk securities. Figure 20 shows the effects of a joint monetary contraction and fiscal deterioration. What is not visible on this plot are the massive increases in debt-to-GDP that result from every monetary policy tightening jointly with deteriorating fiscal backing. Although this model assumes that the home country can inflate away its debt (print the currency in which its nominal debt is denominated), this experiment models the effects of tremendous increases in debt generated by tight monetary policy along with poor fiscal outlook, which is precisely the situation faced by countries like Greece and Spain in the run up to the 2010 sovereign debt crisis. This explains why these countries experienced disproportionately large credit contractions in the wake of policy tightenings.

The U.S., on the other hand, borrows in nominal debt denominated in its own currency. The expectation is that the monetary authority will not permit the central government to default and will accommodate the budgets it sets in advance via seigniorage. This has immediate inflationary impacts as fiscal backing deteriorates, which has expansionary effects on the entire economy, including credit.